Dummit & Foote: Exercises

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Contents

0 Preliminaries

0.1 Basics

For Exercises 1 to 4, let

$$M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

and $\mathscr{B} = \{X \in \mathscr{A} : MX = XM\}$, where $\mathscr{A}$ denotes the set of 2×2 matrices with real entries.

(1) Determine which of the following elements of $\mathscr{A}$ lie in $\mathscr{B}$:

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Solution. Note that the 1st matrix is M itself so that it belongs to $\mathscr{B}$, since $MX = XM = M^2$. Further note that the 3rd, 5th, and 6th matrices are the zero, identity, and exchange matrices respectively. Then the 3rd and 5th belong to $\mathscr{B}$ while the 6th does not. Then only the remaining matrices to check are the 2nd and 4th matrices. For the 2nd matrix:

$$MX = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1\cdot 1 + 1\cdot 1 & 1\cdot 1 + 1\cdot 1 \\ 0\cdot 1 + 1\cdot 1 & 0\cdot 1 + 1\cdot 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}$$
$$XM = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1\cdot 1 + 1\cdot 0 & 1\cdot 1 + 1\cdot 1 \\ 1\cdot 1 + 1\cdot 0 & 1\cdot 1 + 1\cdot 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}$$

Then $MX \neq XM$ and the 2nd matrix does not belong to $\mathscr{B}$. For the 4th matrix:

$$MX = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1\cdot 1 + 1\cdot 1 & 1\cdot 1 + 1\cdot 0 \\ 0\cdot 1 + 1\cdot 1 & 0\cdot 1 + 1\cdot 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$$
$$XM = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1\cdot 1 + 1\cdot 0 & 1\cdot 1 + 1\cdot 1 \\ 1\cdot 1 + 0\cdot 0 & 1\cdot 1 + 0\cdot 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$$

So that $MX \neq XM$. □

(2) Prove that if $P, Q \in \mathscr{B}$, then $P + Q \in \mathscr{B}$ where $+$ denotes the usual sum of two matrices.

Solution. For Exercises 2 and 3, let

$$P = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad Q = \begin{pmatrix} e & f \\ g & h \end{pmatrix}, \quad P + Q = \begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix}, \quad P \cdot Q = \begin{pmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{pmatrix}$$

where $PM = MP$ and $QM = MQ$. Then

$$\begin{aligned} M(P+Q) &= \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix} \\ &= \begin{pmatrix} a+e+c+g & b+f+d+h \\ c+g & d+h \end{pmatrix} \\ &= \begin{pmatrix} a+c & b+d \\ c & d \end{pmatrix} + \begin{pmatrix} e+g & f+h \\ g & h \end{pmatrix} \\ &= MP + MQ \\ &= PM + QM \\ &= \begin{pmatrix} a & a+b \\ c & c+d \end{pmatrix} + \begin{pmatrix} e & e+f \\ g & g+h \end{pmatrix} \\ &= \begin{pmatrix} a+e & a+e+b+f \\ c+g & c+g+d+h \end{pmatrix} \\ &= \begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \\ &= (P+Q)M \end{aligned}$$ □

(3) Prove that if $P, Q \in \mathscr{B}$, then $P \cdot Q \in \mathscr{B}$ where $\cdot$ denotes the usual product of two matrices.

Solution. Using PQ, proceed as we did above with $M(PQ)$. Rewriting the entries will result in $(MP)Q$. Since $P \in \mathscr{B}$, then we have $(PM)Q$. We rewrite entries again to result in $P(MQ)$. Because $Q \in \mathscr{B}$, we have $P(QM)$, and a final rewrite results in $(PQ)M$. □

(4) Find conditions on p, q, r, s which determine precisely when $\begin{pmatrix} p & q \\ r & s \end{pmatrix} \in \mathscr{B}$.

Solution. Let X be the matrix described above. Note that

$$MX = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} p+r & q+s \\ r & s \end{pmatrix}$$
$$XM = \begin{pmatrix} p & q \\ r & s \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p & p+q \\ r & r+s \end{pmatrix}$$

Because $X \in \mathscr{B}$, then we may compare entries to obtain the following:

$$\begin{cases} p+r = p \\ q+s = p+q \\ r = r \\ s = r+s \end{cases}$$

The first and fourth equations force $r = 0$, and the second equation forces $p = s$. Then $\mathscr{B}$ is classified as

$$\mathscr{B} = \left\{ \begin{pmatrix} p & p+q \\ 0 & p \end{pmatrix} \middle| \, p, q \in \mathbb{R} \right\}$$ □

(5) Determine whether the following functions f are well defined:

(a) $f : \mathbb{Q} \to \mathbb{Z}$ defined by $f(a/b) = a$.

(b) $f : \mathbb{Q} \to \mathbb{Q}$ defined by $f(a/b) = a^2/b^2$.

Solution.

(a) No, because $1/2 = 2/4$, but $f(1/2) = 1$ and $f(2/4) = 2$.

(b) Yes; suppose $a/b = c/d$. Then $ad = bc$, or that $a^2d^2 = b^2c^2$. Then $a^2/b^2 = c^2/d^2$, or $f(a/b) = f(c/d)$. □

(6) Determine whether the function $f : R^+ \to \mathbb{Z}$ defined by mapping a real number r to the first digit to the right of the decimal point in a decimal expansion of r is well defined.

Solution. No. Note that $1 = 1.000\ldots = 0.999\ldots$, but $f(1.000\ldots) = 1$ and $f(0.999\ldots) = 9$. □

(7) Let $f : A \to B$ be a surjective map of sets. Prove that the relation

$$a \sim b \iff f(a) = f(b)$$

is an equivalence relation whose equivalence classes are the fibers of f.

Solution. The relation above is an equivalence relation, since $=$ is already an equivalence relation on B.

Consider now some $b \in B$. Since f is surjective, there exists $a \in A$ such that $f(a) = b$. The equivalence class of a is the set $\{x \in A \mid x \sim a\}$. By definition of $\sim$, this is equal to the set $\{x \in A \mid f(x) = f(a) = b\}$, which is precisely the fiber of f over b. □

0.2 Properties of the Integers

1. For each of the following pairs of integers a and b, determine their greatest common divisor, their least common multiple, and write their greatest common divisor in the form $ax + by$ for some integers x and y.

(a) $a = 20, b = 13$
(b) $a = 69, b = 372$
(c) $a = 792, b = 275$
(d) $a = 11391, b = 5673$
(e) $a = 1761, b = 1567$
(f) $a = 507885, b = 60808$

Solution.

(a) Note: for this exercise, we will only do (e) as that has the most steps in calculating both the g.c.d and the Euclidean Algorithm. The l.c.m is obtained by dividing the product ab by (a, b).
$(20, 13) = 1, \text{lcm}(20, 13) = 260, 1 = 2(20) - 3(13)$

(b) $(69, 372) = 3, \text{lcm}(69, 372) = 8556, 27(69) - 5(372)$

(c) $(792, 275) = 11, \text{lcm}(792, 275) = 19800, 8(792) - 23(275)$

(d) $(11391, 5673) = 3, \text{lcm}(11391, 5673) = 21540381, 3 = 253(5673) - 126(11391)$

(e) Applying the Euclidean Algorithm to $a = 1761$ and $b = 1567$, we get:

$$\begin{aligned} 1761 &= (1)1567 + 194 \\ 1567 &= (8)194 + 15 \\ 194 &= (12)15 + 14 \\ 15 &= (1)14 + 1 \end{aligned}$$

Then $(1761, 1567) = 1$ so that $\text{lcm}(1761, 1567) = 2759487$. Reversing the Euclidean Algorithm steps to solve for 1, we get:

$$\begin{aligned} 1 &= 15 - 14 \\ &= 15 - (194 - 12(15)) = 13(15) - 194 \\ &= 13(1567 - 8(194)) - 194 = 13(1567) - 105(194) \\ &= 13(1567) - 105(1761 - 1567) = -105(1761) + 118(1567) \end{aligned}$$

(f) $(507885, 60808) = 691, \text{lcm}(507885, 60808) = 44693880, 691 = 142(60808) - 17(507885)$ □

2. Prove that if the integer k divides the integers a and b then k divides $as + bt$ for every pair of integers s and t.

Solution. Since $k \mid a$ and $k \mid b$, there exists $x, y \in \mathbb{Z}$ such that $a = kx$ and $b = ky$. Then for any $s, t \in \mathbb{Z}$, we have $as + bt = kxs + kyt = k(xs + yt)$ which is divisible by k. □

3. Prove that if n is composite then there are integers a and b such that n divides ab but n does not divide either a or b.

Solution. By the Fundamental Theorem of Arithmetic, n has at least two prime factors a, b such that $1 < a, b < n$. Putting $n = ab$, then $n \mid ab$ but $n \nmid a$ and $n \nmid b$. □

4. Let a, b and N be fixed integers with a and b nonzero and let $d = (a, b)$ be the greatest common divisor of a and b. Suppose x_0 and y_0 are particular solutions to $ax + by = N$ (i.e., $ax_0 + by_0 = N$). Prove for any integer t that the integers

$$x = x_0 + \frac{b}{d}t \text{ and } y = y_0 - \frac{a}{d}t$$

are also solutions to $ax + by = N$ (this is in fact the general solution).

Solution. We have

$$\begin{aligned} ax + by &= a\left(x_0 + \frac{b}{d}t\right) + b\left(y_0 - \frac{a}{d}t\right) \\ &= ax_0 + \frac{ab}{d}t + by_0 - \frac{ab}{d}t \\ &= ax_0 + by_0 = N \end{aligned}$$

□

5. Determine the value $\varphi(n)$ each integer $n \leq 30$ where φ denotes the Euler φ-function.

Solution.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$\varphi(n)$	1	1	2	2	4	2	6	4	6	4	10	4	12	6	8

n	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
$\varphi(n)$	8	16	6	18	8	12	10	22	8	20	12	18	12	28	8

□

6. Prove the Well Ordering Property of $\mathbb{Z}$ by induction and prove the minimal element is unique.

Solution. Let $A \subseteq \mathbb{N}$ be nonempty. If $0 \in A$, then it has a minimal element. Suppose now that $0 \notin A$. Moreover, suppose that $1, 2, \ldots, k \notin A$ for some k. By strong induction, it follows that $k+1 \notin A$. However, this results in there being no positive integer in A, contradicting that it is nonempty, thus it must have a minimal element. Moreover, if it had two minimal elements a, b, then it follows that $a \leq b$ and $b \leq a$ by definition of minimal element. Hence, $a = b$, and the minimal element of A is unique. □

7. If p is a prime, prove that there do not exist nonzero integers a and b such that $a^2 = pb^2$ (i.e., $\sqrt{p}$ is not a rational number).

Solution. Suppose $\sqrt{p}$ is a rational number. Then there exists $a, b \in \mathbb{Z}$ such that $a^2 = b^2 p$, and $(a, b) = 1$. If $(a, b) = d \neq 1$, then take $a' = a/d$ and $b' = b/d$ instead. Then $p \mid a^2 = a \cdot a$, and $p \mid a$. We then have $a = kp$ for some $k \in \mathbb{Z}$. Then $b^2 = k^2 p$ so that $p \mid b$, and $b = mp$ for $m \in \mathbb{Z}$. This contradicts that $(a, b) = 1$, hence $\sqrt{p}$ cannot be a rational number. □

8. Let p be a prime, $n \in \mathbb{N}$. Find a formula for the largest power of p which divides $n! = n(n-1)(n-2)\ldots 2 \cdot 1$ (it involves the greatest integer function).

Solution. Note that in $n!$, there are $\lfloor n/p \rfloor$ integers that are divisible by p, where the greatest integer function is necessary since n may not be a perfect multiple of p (consider $n = 36$ and $p = 7$. Then there are the multiples $7, 14, 21, 28$, and 35 which is $\lfloor 36/7 \rfloor = 5$). However, this alone does not account for the factors of p in $n!$, so we move onto p^2. In this case, there are $\lfloor n/p^2 \rfloor$ integers divisible by p^2. We continue this process until a certain integer $a \in \mathbb{N}$ such that $p^a \leq n$. (We must bound p^a above by n, up to equality, since n may be a power of p, and any power a such that $p^a > n$ would result in $\lfloor n/p^a \rfloor = 0$). It then follows that $a \leq \log_p(n)$, or that the maximum power of p (and any of its multiples) that divide $n!$ is given by $a = \lfloor \log_p(n) \rfloor$. Thus, the largest power of p that divides $n!$ is given by

$$\sum_{i=1}^{\lfloor \log_p(n) \rfloor} \left\lfloor \frac{n}{p^i} \right\rfloor$$

□

9. Write a computer program to determine the greatest common divisor (a, b) of two integers a and b and to express (a, b) in the form $ax + by$ for some integers x and y.

Require: Integers a, b (not both zero)
Ensure: Integers g, x, y such that $g = (a, b)$ and $ax + by = g$

$x_1 \leftarrow 1$, $y_1 \leftarrow 0$
$x_2 \leftarrow 0$, $y_2 \leftarrow 1$
while $b \neq 0$ **do**
  $q \leftarrow 0$
  $r \leftarrow a$
  while $(r \geq b)$ or $(r \leq -b)$ **do**
    $r \leftarrow r - b$
    $q \leftarrow q + 1$
  end while
  $a \leftarrow b$
  $b \leftarrow r$

$\quad (x_1, x_2) \leftarrow (x_2, x_1 - q \times x_2)$
$\quad (y_1, y_2) \leftarrow (y_2, y_1 - q \times y_2)$
end while
$g \leftarrow a$
return (g, x_1, y_1)

10. Prove for any given positive integer N there exist only finitely many integers n with $\varphi(n) = N$ where φ denotes Euler's φ-function. Conclude in particular that φ tends to infinity as n tends to infinity.

Solution. Let $N \in \mathbb{N}$ be fixed, and let $n \in \mathbb{N}$ such that $\varphi(n) = N$. We can break apart n into its primes p_i (where $p_1 < p_2 < \dots$) with associated exponents α_i:

$$n = \prod_{i=1}^{k} p_i^{\alpha_i} \implies \varphi(n) = \prod_{i=1}^{k} p_i^{\alpha_i - 1}(p_i - 1) = N$$

In particular, each of the terms $p_i - 1$ divides N so that $p_i - 1 \leq N$, or $p_i \leq N + 1$. It follows that any prime q of some n must be such a prime where $q - 1 \leq N$. Moreover, for some $p_i^{\alpha_i}$ that is a part of n's prime factorization, it follows that $p_i^{\alpha_i - 1}$ divides N. Since $\alpha_i - 1$ is finite, this limits the exponents for any particular p_i. With a finite list of primes and finitely many exponents, it follows that the amount of n such that $\varphi(n) = N$ is finite.

For some $N_0 \in \mathbb{N}$, there are finitely many n such that $\varphi(n) = N_0$. Then there are infinitely many $m > n$ such that $\varphi(m) > N_0$, or that $\varphi(n)$ tends towards infinity. □

11. Prove that if d divides n then $\varphi(d)$ divides $\varphi(n)$ where φ denotes Euler's φ-function.

Solution. Let $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ with $d \mid n$. Then d is a composition of some p_i present in n, so some α_i may go to 0 or are less. In particular, we have that $d = p_1^{\beta_1} p_2^{\beta_2} \cdots p_k^{\beta_k}$, where $\beta_i \leq \alpha_i$ for all i. To see if $\varphi(d) \mid \varphi(n)$, it follows to check if $p_i^{\beta_i - 1}(p_i - 1)$ divides $p_i^{\alpha_i - 1}(p_i - 1)$, which further simplifies to checking if $p_i^{\beta_i}$ divides $p_i^{\alpha_i}$ for each i. Clearly, $p_i^{\alpha_i} = p_i^{\beta_i} \cdot p_i^{\alpha i - \beta_i}$, hence $\varphi(d) \mid \varphi(n)$. □

0.3 $\mathbb{Z}/n\mathbb{Z}$: The Integers Modulo n

1. Write down explicitly all the elements in the residue classes of $\mathbb{Z}/18\mathbb{Z}$.

Solution.

$$\begin{aligned}
\bar{0} &= \{0 + 18k : k \in \mathbb{Z}\} = \{0, 18, -18, 36, -36, \dots\} \\
\bar{1} &= \{1 + 18k : k \in \mathbb{Z}\} = \{1, 19, -17, 37, -35, \dots\} \\
\bar{2} &= \{2 + 18k : k \in \mathbb{Z}\} = \{2, 20, -16, 38, -34, \dots\} \\
\bar{3} &= \{3 + 18k : k \in \mathbb{Z}\} = \{3, 21, -15, 39, -33, \dots\} \\
\bar{4} &= \{4 + 18k : k \in \mathbb{Z}\} = \{4, 22, -14, 40, -32, \dots\} \\
\bar{5} &= \{5 + 18k : k \in \mathbb{Z}\} = \{5, 23, -13, 41, -31, \dots\} \\
\bar{6} &= \{6 + 18k : k \in \mathbb{Z}\} = \{6, 24, -12, 42, -30, \dots\} \\
\bar{7} &= \{7 + 18k : k \in \mathbb{Z}\} = \{7, 25, -11, 43, -29, \dots\} \\
\bar{8} &= \{8 + 18k : k \in \mathbb{Z}\} = \{8, 26, -10, 44, -28, \dots\} \\
\bar{9} &= \{9 + 18k : k \in \mathbb{Z}\} = \{9, 27, -9, 45, -27, \dots\} \\
\overline{10} &= \{10 + 18k : k \in \mathbb{Z}\} = \{10, 28, -8, 46, -26, \dots\} \\
\overline{11} &= \{11 + 18k : k \in \mathbb{Z}\} = \{11, 29, -7, 47, -25, \dots\} \\
\overline{12} &= \{12 + 18k : k \in \mathbb{Z}\} = \{12, 30, -6, 48, -24, \dots\} \\
\overline{13} &= \{13 + 18k : k \in \mathbb{Z}\} = \{13, 31, -5, 49, -23, \dots\} \\
\overline{14} &= \{14 + 18k : k \in \mathbb{Z}\} = \{14, 32, -4, 50, -22, \dots\} \\
\overline{15} &= \{15 + 18k : k \in \mathbb{Z}\} = \{15, 33, -3, 51, -21, \dots\} \\
\overline{16} &= \{16 + 18k : k \in \mathbb{Z}\} = \{16, 34, -2, 52, -20, \dots\} \\
\overline{17} &= \{17 + 18k : k \in \mathbb{Z}\} = \{17, 35, -1, 53, -19, \dots\}
\end{aligned}$$

$\square$

2. Prove that the distinct equivalence classes in $\mathbb{Z}/n\mathbb{Z}$ are precisely $\bar{0}, \bar{1}, \bar{2}, \ldots, \overline{n-1}$ (use the Division Algorithm).

   Solution. Clearly, $\bar{0}, \bar{1}, \bar{2}, \ldots, \overline{n-1} \in \mathbb{Z}/n\mathbb{Z}$. Let $\bar{a} \in \mathbb{Z}/n\mathbb{Z}$. By the Division Algorithm, there exists $q \in \mathbb{Z}$ and $0 \le r < n$ such that $a = nq + r$. Then $a \equiv r \bmod n$, so that $\bar{a} = \bar{r}$, where r is any one of the aforementioned equivalence classes. Moreover, if $\bar{a} = \bar{b}$ where $0 \le a, b < n$, it follows that $n \mid (a-b)$ so that $a - b = 0$, or $a = b$. Hence, the equivalence classes of $\mathbb{Z}/n\mathbb{Z}$ are precisely the set above. $\square$

3. Prove that if $a = a_n 10^n + a_{n-1} 10^{n-1} + \cdots + a_1 10 + a_0$ is any positive integer then $a \equiv (a_n + a_{n-1} + \cdots + a_1 + a_0) \bmod 9$ (note that this is the usual arithmetic rule that the remainder after division by 9 is the same as the sum of the decimal digits mod 9—in particular an integer is divisible by 9 if and only if the sum of its digits is divisible by 9) [note that $10 \equiv 1 \bmod 9$].

   Solution. Using the note, then

$$\begin{aligned} a &\equiv (a_n 10^n + a_{n-1} 10^{n-1} + \cdots + a_1 10 + a_0) \bmod 9 \\ &\equiv (a_n 1^n + a_{n-1} 1^{n-1} + \cdots + a_1 1 + a_0) \bmod 9 \\ &= (a_n + a_{n-1} + \cdots + a_1 + a_0) \bmod 9 \end{aligned}$$

$\square$

4. Compute the remainder when 37^{100} is divided by 29.

   Solution. Note the following:

$$\begin{aligned} 37^2 &\equiv 6 \bmod 29 \\ 37^4 &\equiv 6^2 \bmod 29 \equiv 7 \bmod 29 \\ 37^8 &\equiv 7^2 \bmod 29 \equiv 20 \bmod 29 \\ 37^{16} &\equiv 20^2 \bmod 29 \equiv 23 \bmod 29 \equiv -6 \bmod 29 \\ 37^{32} &\equiv (-6)^2 \bmod 29 \equiv 7 \bmod 29 \\ 37^{64} &\equiv 20 \bmod 29 \end{aligned}$$

   Then we have

$$37^{64} 37^{32} 37^4 \equiv 20 \cdot 7 \cdot 7 \bmod 29 \equiv 20^2 \bmod 29 \equiv 23 \bmod 29$$

   Hence the remainder when dividing 37^{100} by 29 is 23. $\square$

5. Compute the last two digits of 9^{1500}.

   Solution. Note that $9^3 \equiv 29 \bmod 100$ so that $29^3 \equiv 89 \bmod 100$. Then $89 \cdot 9 \equiv 1 \bmod 100$, or $9^{10} \equiv 1 \bmod 100$. Since 1500 is a multiple of 10, then $9^{1500} = (9^{10})^{150} \equiv 1^{150} \bmod 100 \equiv 1 \bmod 100$. Then the last two digits is 01. $\square$

6. Prove that the square of the elements in $\mathbb{Z}/4\mathbb{Z}$ are just $\bar{0}$ and $\bar{1}$.

   Solution.

$$\begin{aligned} 0^2 &= 0 \equiv 0 \bmod 4 \\ 1^2 &= 1 \equiv 1 \bmod 4 \\ 2^2 &= 4 \equiv 0 \bmod 4 \\ 3^2 &= 9 \equiv 1 \bmod 4 \end{aligned}$$

$\square$

7. Prove that for any integers a and b that $a^2 + b^2$ never leaves a remainder of 3 when divided by 4 (use the previous exercise).

Solution. Since a^2 and b^2 is either $0 \bmod 4$ or $1 \bmod 4$, then we have 4 potential sums:

$$\begin{aligned} 0+0 &\equiv 0 \bmod 4 \\ 0+1 &\equiv 1 \bmod 4 \\ 1+0 &\equiv 1 \bmod 4 \\ 1+1 &\equiv 2 \bmod 4 \end{aligned}$$

In any of the sums, there is no remainder of 3 when dividing by 4. □

8. Prove that the equation $a^2+b^2=3c^2$ has no solutions in nonzero integers a, b and c. [Consider the equation mod4 as in the previous two exercises and show that a, b and c would all have to be divisible by 2. Then each of a^2, b^2 and c^2 has a factor of 4 and by dividing through by 4 show that there would be a smaller set of solutions to the original equation. Iterate to reach a contradiction.]

   Solution. As hinted, consider the equation in $\bmod 4$. The left side must be 0 or 1, while the right side must be 0 or 3, hence both sides must be equivalent to $0 \bmod 4$. Then $3c^2 \equiv 0 \bmod 4$ implies c is even. If a is even, then b is even. If a is odd, then b must be odd. However, putting $a=2x+1$ and $b=2y+1$ for $x, y \in \mathbb{Z}$ results in $a^2+b^2=(2x+1)^2+(2y+1)^2 \equiv 2 \bmod 4$, contradicting that $a^2+b^2 \equiv 0 \bmod 4$. It must be that a and b are even. We may then divide the original equation by 4 to get a new equation $r^2+s^2=3t^2$, where $r<a, s<b, t<c$. But then we may use the same reasoning to conclude that the new r, s, t must also be even, resulting in a new equation with smaller integer solutions. This process cannot be iterated indefinitely as we cannot halve any integer indefinitely and remain an integer. Hence, the original equation does not have integer solutions. □

9. Prove that the square of any odd integer always leaves a remainder of 1 when divided by 8.

   Solution. Take $a=2k+1, k \in \mathbb{Z}$. Then $a^2=4k^2+4k+1=4k(k+1)+1$. Note that $k(k+1)$ is even, so that $4k(k+1)$ is divisible by 8. Then $a^2 \equiv 1 \bmod 8$. □

10. Prove that the number of elements of $(\mathbb{Z}/n\mathbb{Z})^\times$ is $\varphi(n)$ where φ denotes the Euler φ-function.

    Solution. Recall that $\varphi(n)$ produces the number of integers that are relatively prime to n. It suffices to prove that the elements of $(\mathbb{Z}/n\mathbb{Z})^\times$ are the equivalence classes of $\mathbb{Z}/n\mathbb{Z}$ whose representatives are relatively prime to n.

    Let $\overline{a} \in (\mathbb{Z}/n\mathbb{Z})^\times$. Then there exists $\overline{b} \in (\mathbb{Z}/n\mathbb{Z})^\times$ such that $\overline{a} \cdot \overline{b} = \overline{1}$. In particular, $\overline{a} \cdot \overline{b} - \overline{1} = \overline{0}$, or $n \mid (ab-1)$. Then $nx+ab=1$ for $x \in \mathbb{Z}$, which shows that $(a,n)=1$. Conversely, if $\overline{a} \in (\mathbb{Z}/n\mathbb{Z})^\times$ such that $(a,n)=1$, then there exists $b, x \in \mathbb{Z}$ such that $ab+xn=1$, or $ab=1 \bmod n$. It follows that $\overline{b}$ is the inverse of $\overline{a} \in (\mathbb{Z}/n\mathbb{Z})^\times$. □

11. Prove that if $\overline{a}, \overline{b} \in (\mathbb{Z}/n\mathbb{Z})^\times$, then $\overline{a} \cdot \overline{b} \in (\mathbb{Z}/n\mathbb{Z})^\times$.

    Solution. It follows that there exist $\overline{x}, \overline{y} \in (\mathbb{Z}/n\mathbb{Z})^\times$ such that $\overline{a} \cdot \overline{x} = \overline{b} \cdot \overline{y} = \overline{1}$. In particular:

    $$\overline{1} = \overline{1} \cdot \overline{1} = (\overline{a} \cdot \overline{x}) \cdot (\overline{b} \cdot \overline{y}) = (\overline{a} \cdot \overline{b}) \cdot (\overline{x} \cdot \overline{y})$$

    So that the multiplicative inverse of $\overline{a} \cdot \overline{b}$ is $\overline{x} \cdot \overline{y}$. □

12. Let $n \in \mathbb{Z}, n>1$ and let $a \in \mathbb{Z}$ with $1 \le a \le n$. Prove if a and n are not relatively prime, there exists an integer b with $1 \le b < n$ such that $ab \equiv 0 \bmod n$ and deduce that there cannot be an integer c such that $ac \equiv 1 \bmod n$.

    Solution. Since a and n are not relatively prime, then $(a,n)=d$ where $1<d \le a$. Then $n/d, a/d \in \mathbb{Z}$, and $a(n/d)=n(a/d) \equiv 0 \bmod n$ so that $b=n/d$. Moreover, if there was such a c such that $ac \equiv 1 \bmod n$, then $abc \equiv b \bmod n$, which is false since $ab \equiv 0 \bmod n$. Hence, no such b may exist. □

13. Let $n \in \mathbb{Z}, n>1$ and let $a \in \mathbb{Z}$ with $1 \le a \le n$. Prove that if a and n are relatively prime then there is an integer c such that $ac \equiv 1 \bmod n$ [use the fact that the g.c.d. of two integers is a $\mathbb{Z}$-linear combination of the integers].

    Solution. Since $(a,n)=1$, there exists $c, x \in \mathbb{Z}$ such that $ac+nx=1$. Then $ac \equiv 1 \bmod n$. □

14. Conclude from the previous two exercises that $(\mathbb{Z}/n\mathbb{Z})^\times$ is the set of elements $\overline{a}$ of $\mathbb{Z}/n\mathbb{Z}$ with $(a,n)=1$ and hence prove Proposition 4. Verify this directly in the case $n=12$.

Solution. The previous two exercises show that a and n are relatively prime if and only if there exists b such that $ab \equiv 1 \bmod n$, which is exactly the proposition. For $n = 12$, the elements $1, 5, 7, 11$ are relatively prime to 12, so that $(\mathbb{Z}/11\mathbb{Z})^\times = \{\overline{1}, \overline{5}, \overline{7}, \overline{11}\}$ whose inverses are $\overline{1}, \overline{7}, \overline{5}, \overline{11}$ respectively. □

15. For each of the following pairs of integers a and n, show that a is relatively prime to n and determine the multiplicative inverse of $\overline{a}$ in $\mathbb{Z}/n\mathbb{Z}$.
    (a) $a = 13, n = 20$
    (b) $a = 69, n = 89$
    (c) $a = 1891, n = 3797$
    (d) $a = 6003722857, n = 77695236973$

    Solution.

    (a) Refer to the previous set of exercises on how to do the Euclidean Algorithm and how to calculate such an x such that $ax + by = 1$, or $ax \equiv 1 \bmod n$. For this exercise, we obtain $-3 \equiv 17 \bmod 20$, so the inverse is $\overline{17}$.

    (b) $40(69) - 31(89) = 1 \implies 40 \cdot 89 \equiv 1 \bmod 89$, so the inverse is $\overline{40}$.

    (c) $253(1891) - 126(3797) = 1 \implies 253 \cdot 1891 \equiv 1 \bmod 3797$, so the inverse is $\overline{253}$.

    (d) $17n - 220a = 1 \implies -220a \equiv 1 \bmod n \implies 77695237193a \equiv 1 \bmod n$, so the inverse is $\overline{77695237193}$. □

16. Write a computer program to add and multiply mod n, for any n given as input. The output of these operations should be the least residues of the sums and products of two integers. Also include the feature that if $(a, n) = 1$, an integer c between 1 and $n - 1$ such that $\overline{a} \cdot \overline{c} = \overline{1}$ may be printed on request. (Your program should not, of course, simply quote "mod" functions already built into many systems).

1 Introduction to Groups

1.1 Basic Axioms and Examples

Let G be a group.

(1) Determine which of the following binary operations are associative:

(a) the operation $\star$ on $\mathbb{Z}$ defined by $a \star b = a - b$

(b) the operation $\star$ on $\mathbb{R}$ defined by $a \star b = a + b + ab$

(c) the operation $\star$ on $\mathbb{Q}$ defined by $a \star b = \dfrac{a+b}{5}$

(d) the operation $\star$ on $\mathbb{Z} \times \mathbb{Z}$ defined by

$$(a,b) \star (c,d) = (ad + bc,\, bd)$$

(e) the operation $\star$ on $\mathbb{Q} - \{0\}$ defined by $a \star b = \dfrac{a}{b}$

Solution.

(a) Not associative: $(2-0)-2 \neq 2-(0-2)$.

(b) Associative:

$$\begin{aligned}(a \star b) \star c &= (a+b+ab) \star c \\ &= (a+b+ab) + c + (a+b+ab)c \\ &= a+b+ab+c+ac+bc+abc \\ &= a+b+c+bc+ab+ac+abc \\ &= a+(b+c+bc)+a(b+c+bc) \\ &= a \star (b+c+bc) \\ &= a \star (b \star c)\end{aligned}$$

(c) Not associative: $(1 \star 0) \star 2 = 11/25, 1 \star (0 \star 2) = 7/25$.

(d) Associative:

$$\begin{aligned}[(a,b)\star(c,d)]\star(e,f) &= (ad+bc, bd)\star(e,f) \\ &= ((ad+bc)f + bde, bdf) \\ &= (adf + bcf + bde, bdf) \\ &= (adf + b(cf+de), bdf) \\ &= (a,b)\star(cf+de, df) \\ &= (a,b)\star[(c,d)\star(e,f)]\end{aligned}$$

(e) Not associative: $(1 \star 2) \star 3 = 1/6, 1 \star (2 \star 3) = 3/2$. $\square$

(2) Decide which of the binary operations in the preceding exercise are commutative.

Solution.

(a) Not commutative: $1 - 0 \neq 0 - 1$.

(b) Commutative: $a \star b = a + b + ab = b + a + ba = b \star a$.

(c) Commutative: $a \star b = (a+b)/5 = (b+a)/5 = b \star a$.

(d) Commutative: $(a,b)\star(c,d) = (ad+bc, bd) = (cb+da, db) = (c,d)\star(a,b)$.

(e) Not commutative: $1/2 \neq 2/1$. $\square$

(3) Prove that addition of residue classes in $\mathbb{Z}/n\mathbb{Z}$ is associative (you may assume it is well defined).

Solution. Let $\overline{a}, \overline{b}, \overline{c} \in \mathbb{Z}/n\mathbb{Z}$. Then $(\overline{a}+\overline{b})+\overline{c} = \overline{a+b}+\overline{c} = \overline{(a+b)+c} = \overline{a+(b+c)} = \overline{a}+\overline{b+c} = \overline{a}+(\overline{b}+\overline{c})$. $\square$

(4) Prove that multiplication of residue classes in $\mathbb{Z}/n\mathbb{Z}$ is associative (you may assume it is well defined).

Solution. Uses similar reasoning as the previous exercise (where $\cdot$ is associative over $\mathbb{Z}$). $\square$

(5) Prove for all $n > 1$ that $\mathbb{Z}/n\mathbb{Z}$ is not a group under multiplication of residue classes.

Solution. Note that in any $\mathbb{Z}/n\mathbb{Z}$, $\bar{1}$ is a multiplicative identity. Moreover, there is no $x \in \mathbb{Z}$ such that $0x = 1$ so that $\bar{0} \in \mathbb{Z}/n\mathbb{Z}$ has no multiplicative inverse. $\square$

(6) Determine which of the following sets are groups under addition:

(a) the set of rational numbers (including $0 = 0/1$) in lowest terms whose denominators are odd
(b) the set of rational numbers (including $0 = 0/1$) in lowest terms whose denominators are even
(c) the set of rational numbers of absolute value < 1
(d) the set of rational numbers of absolute value ≥ 1 together with 0
(e) the set of rational numbers with denominators equal to 1 or 2
(f) the set of rational numbers with denominators equal to 1, 2, or 3

Solution. Let S denote the set in each part.

(a) Clearly, $0 \in S$ and for any $r \in S$, then $-r \in S$ is the additive inverse. To prove closure, let $a/b, c/d \in S$, and consider its sum $(ad+bc)/bd$. Since b, d are odd, then bd is also odd. Since having an even factor results in an even number, then any divisor of bd must also be odd. Then $(ad+bc, bd)$ must also be odd, and if we divide both the numerator and the denominator by this quantity, then we still result in odd numbers. Hence, A is closed under addition and is a group.
(b) S is not a group, since $1/2 \in S$, but $1/2 + 1/2 = 2/2 = 1/1 \notin S$.
(c) Same as previous.
(d) S is not a group, since $3/2, 1 \in S$, but $3/2 - 1 = 1/2 \notin S$.
(e) Using the Division Algorithm, any $p \in S$ may be split into the form $a + b/2$, where a is even and b is either 0 (if the numerator p is even) or 1 (if the numerator of p is odd). Then if we have $a_1 + b_1/2, a_2 + b_2/2 \in S$ and we consider their sum, then either both b_i is 0 or 1, in which case we remain an integer with denominator 1, or just one of the b_i is 1 in which case we end with a rational with denominator 2 (WLOG, suppose $b_1 = 1$ and $b_2 = 0$. Then we get the rational $(2a_1 + 2a_2 + 1)/2$). Hence, S is closed under addition, and it also has identity 0 and additive inverse $-p$ so that S is a group.
(f) S is not a group, since $1/2, 1/3 \in S$, but $1/2 + 1/3 = 5/6 \notin S$. $\square$

(7) Let $G = \{x \in \mathbb{R} \mid 0 \leq x < 1\}$ and for $x, y \in G$ let $x \star y$ be the fractional part of $x + y$ (i.e., $x \star y = x + y - [x + y]$ where $[a]$ is the greatest integer less than or equal to a). Prove that $\star$ is a well defined binary operation on G and that G is an abelian group under $\star$ (called the *real numbers mod I*).

Solution. Let $x, y \in G$. Then we have two cases:

- $0 \leq x + y < 1$: Then $[x + y] = 0$, and $x \star y = x + y \in G$,
- $1 \leq x + y < 2$: Then $[x + y] = 1$, and $x \star y = x + y - 1 \in G$.

so that $\star$ is well defined on G. To show associativity, let $z \in G$. Then there are three cases:

- If $0 \leq x + y, y + z < 1$, then $[x + y] = [y + z] = 0$:

$$\begin{aligned}(x \star y) \star z &= (x + y - [x + y]) \star z \\ &= x + y + z - [x + y + z] \\ &= x \star (y + z - [y + z]) \\ &= x \star (y \star z)\end{aligned}$$

- If $1 \leq x + y, y + z < 2$, then $[x + y] = [y + z] = 1$:

$$\begin{aligned}(x \star y) \star z &= (x + y - [x + y]) \star z \\ &= (x + y - 1) + z - [x + y - 1 + z] \\ &= x + (y + z - 1) - [x + y + z - 1] \\ &= x \star (y + z - 1) - [x + (y + z - 1)] \\ &= x \star (y + z - [y + z]) \\ &= x \star (y \star z)\end{aligned}$$

- If $1 \le x+y < 2$ and $0 \le y+z < 1$, then $[x+y]=1$ and $[y+z]=0$. The case of $0 \le x+y < 1$ and $1 \le y+z < 2$ is similar:

$$\begin{aligned}(x\star y)\star z &= (x+y-[x+y])\star z\\ &= (x+y-1)+z-[x+y-1+z]\\ &= x+y-1+z-([x+y+z]-1)\\ &= x+y+z-1-[x+y+z]+1\\ &= x+y+z-[x+y+z]\\ &= x\star(y+z-[y+z])\\ &= x\star(y\star z)\end{aligned}$$

Then $\star$ is associative. Moreover, commutativity of regular addition implies commutativity of G. For the identity element, note that $0 \in G$ and $0\star x = x+0-[x]=x$. To determine the inverse of some $x \in G$, suppose $w \in G$ such that $x\star w = 0$. Then $x+w-[x+w]=0$. Note that $0 \le x+w < 2$, so $[x+w]=0$ or 1. If $[x+w]=0$, then $w=-x$. Since $w, x \in G$, then this is when $x=w=0$. If $[x+w]=1$, then $w=1-x$ for $0<x<1$. Hence, G is closed under inverses and is a group. □

(8) Let $G=\{z\in\mathbb{C}: z^n=1 \text{ for some } n\in\mathbb{Z}^+\}$.

(a) Prove that G is a group under multiplication (called the group of roots of unity in $\mathbb{C}$).

(b) Prove that G is not a group under addition.

Solution.

(a) Note that $1\in G$, so G has an identity. Moreover, if $zz^{-1}=1$ for some $z\in G$, then $z^{-1}=1/z\in G$ because $(1/z)^n = 1/z^n = 1$. Lastly, let $z, w\in G$, where $z^n=1$ and $w^m=1$ for $n,m\in\mathbb{Z}^+$. Then

$$(zw)^{mn} = (z^n)^m(w^m)^n = 1$$

so that $zw\in G$. Moreover, multiplication is associative in G since $G\subseteq\mathbb{C}$.

(b) Note that $1,-1\in G$, but $1+(-1)=0\notin G$, since there is no n such that $0^n=1$. □

(9) Let $G=\{a+b\sqrt{2}\in\mathbb{R}: a,b\in\mathbb{Q}\}$.

(a) Prove that G is a group under addition.

(b) Prove that the nonzero elements of G are a group under multiplication. ["Rationalize the denominators" to find multiplicative inverses.]

Solution.

(a) Since $0+0\sqrt{2}=0\in G$, then G has an identity. Moreover, for any $a+b\sqrt{2}\in G$, it has inverse $-a-b\sqrt{2}\in G$. Associativity in G follows from associativity in $\mathbb{R}$, and for $a+b\sqrt{2}, c+d\sqrt{2}\in G$, we have $(a+b\sqrt{2})+(c+d\sqrt{2})=(a+c)+(b+d)\sqrt{2}$, where $a+b, c+d\in q$ so that it is closed under addition. Hence, G is a group.

(b) Since $1+0\sqrt{2}=1\in G$, then G has identity. Moreover, for $a+b\sqrt{2}$, we take:

$$\frac{1}{a+b\sqrt{2}} = \frac{1}{a+b\sqrt{2}}\cdot\frac{a-b\sqrt{2}}{a-b\sqrt{2}} = \frac{a-b\sqrt{2}}{a^2-2b^2} = \frac{a}{a^2-2b^2} - \frac{b}{a^2-2b^2}\sqrt{2}$$

where $a^2-2b^2\neq 0$ (otherwise, then $a/b=\pm\sqrt{2}$, which is impossible). Then G is closed under inverses. Associativity in G follows from associativity of multiplication in $\mathbb{R}$, and for $a+b\sqrt{2}, c+d\sqrt{2}\in G$, we have $(a+b\sqrt{2})(c+d\sqrt{2})=(ac+2bd)+(ad+bc)\sqrt{2}$ where $ac+2bd, ad+bc\in\mathbb{Q}$ so that G is closed under multiplication. Then G is a group. □

(10) Prove that a finite group is abelian if and only if its group table is a symmetric matrix.

Solution. Let G be a finite group with elements $g_1, g_2, \ldots, g_n$ for finite n and $g_1=1$. Then G is abelian if and only if $g_ig_j = g_jg_i$ for any i, j if and only if the group table is a symmetric matrix. □

(11) Find the orders of each element of the additive group $\mathbb{Z}/12\mathbb{Z}$.

Solution. For each $\overline{x} \in \mathbb{Z}/12\mathbb{Z}$, add it to itself until we arrive at $\overline{0}$. $|\overline{0}| = 1$, and $\overline{1} = 12$ since $12 \cdot \overline{1} = \overline{12} = \overline{0}$. In particular, we have:

$\overline{x}$	$\overline{0}$	$\overline{1}$	$\overline{2}$	$\overline{3}$	$\overline{4}$	$\overline{5}$	$\overline{6}$	$\overline{7}$	$\overline{8}$	$\overline{9}$	$\overline{10}$	$\overline{11}$
$\lvert\overline{x}\rvert$	1	12	6	4	3	12	2	12	3	4	6	12

□

(12) Find the orders of the following elements of the multiplicative group $(\mathbb{Z}/12\mathbb{Z})^\times$: $\overline{1}, \overline{-1}, \overline{5}, \overline{7}, \overline{-7}, \overline{13}$.

Solution.

$\overline{x}$	$\overline{1}$	$\overline{-1}$	$\overline{5}$	$\overline{7}$	$\overline{-7}$	$\overline{13}$
$\lvert\overline{x}\rvert$	1	2	2	2	2	1

□

(13) Find the orders of the following elements of the additive group $\mathbb{Z}/36\mathbb{Z}$: $\overline{1}, \overline{2}, \overline{6}, \overline{9}, \overline{10}, \overline{12}, \overline{-1}, \overline{-10}, \overline{-18}$.

Solution.

$\overline{x}$	$\overline{1}$	$\overline{2}$	$\overline{6}$	$\overline{9}$	$\overline{10}$	$\overline{12}$	$\overline{-1}$	$\overline{-10}$	$\overline{-18}$
$\lvert\overline{x}\rvert$	36	18	6	4	18	3	36	18	2

□

(14) Find the orders of the following elements of the multiplicative group $(\mathbb{Z}/36\mathbb{Z})^\times$: $\overline{1}, \overline{-1}, \overline{5}, \overline{13}, \overline{-13}, \overline{17}$.

Solution.

$\overline{x}$	$\overline{1}$	$\overline{-1}$	$\overline{5}$	$\overline{13}$	$\overline{-13}$	$\overline{17}$
$\lvert\overline{x}\rvert$	1	2	6	3	6	2

□

(15) Prove that $(a_1a_2\dots a_n)^{-1} = a_n^{-1}a_{n-1}^{-1}\dots a_1^{-1}$ for all $a_1, a_2, \dots, a_n \in G$.

Solution. For $n = 1$, the result follows. Suppose it holds for $n = k \in \mathbb{Z}^+$. For $n = k+1$, we have:

$$\begin{aligned}(a_1a_2\dots a_ka_{k+1})^{-1}(a_{k+1}^{-1}a_k^{-1}\dots a_2^{-1}a_1^{-1}) &= (a_1a_2\dots a_k)(a_{k+1}a_{k+1}^{-1})(a_k^{-1}\dots a_2^{-1}a_1^{-1}) \\ &= (a_1a_2\dots a_k)^{-1}(a_k^{-1}\dots a_2^{-1}a_1^{-1}) \\ &= 1\end{aligned}$$

The result follows by induction. □

(16) Let x be an element of G. Prove that $x^2 = 1$ if and only if $|x|$ is either 1 or 2.

Solution. If $x^2 = 1$, then $|x| \le 2$. Then $|x| = 1$ or 2. If $|x| = 1$, then $x = 1$, and $x^2 = 1^2 = 1$. If $|x| = 2$, then $x^2 = 1$. □

(17) Let x be an element of G. Prove that if $|x| = n$ for some positive integer n, then $x^{-1} = x^{n-1}$.

Solution. Note that $x^n = x^{n-1}x = 1$. Then $x^{n-1} = x^{-1}$. □

(18) Let x, y be elements of G. Prove that $xy = yx$ if and only if $y^{-1}xy = x$ if and only if $x^{-1}y^{-1}xy = 1$.

Solution.

$$\begin{aligned}xy &= yx \\ y^{-1}xy &= y^{-1}yx \\ y^{-1}xy &= x \\ x^{-1}y^{-1}xy &= x^{-1}x \\ x^{-1}y^{-1}xy &= 1\end{aligned}$$

□

(19) Let $x \in G$ and let $a, b \in \mathbb{Z}^+$.

(a) Prove that $x^{a+b} = x^a x^b$ and $(x^a)^b = x^{ab}$.

(b) Prove that $(x^a)^{-1} = x^{-a}$.

(c) Establish part (a) for arbitrary integers a and b (positive, negative, or zero).

Solution.

(a) Since x^a has a amount of x and x^b has b amount of x, and x^{a+b} has $a+b$ amount of x, the result follows. Similarly, b amount of x^a means there are ab amount of x, or x^{ab}.

(b) Note that $x^{-a} = (x^{-1})^a$. For $a = 1$, this follows. Suppose $(x^a)^{-1} = (x^{-1})^a$ for $a = n \in \mathbb{Z}^+$. Then for $a = n+1$, we have $(x^{-1})^{n+1}(x^{n+1}) = (x^{-1})^n x^{-1} x x^n = 1$, which is true by the inductive hypothesis.

(c) The case where both a and b are positive is done. If $a \in \mathbb{Z}$ and $b = 0$, then $x^{a+0} = x^a x^0 = x^a$. Moreover, we have $(x^a)^0 = 1$ by definition, so $x^{a\cdot 0} = x^0 = 1$. If $a > 0$ and $b < 0$, we have two cases for the first part of (a):

- If $a + b > 0$, then $x^a = x^{a+b-b} = x^{a+b} x^{-b}$. Then $x^a x^b = x^{a+b}$.
- If $a + b < 0$, then $x^b = (x^{-b})^{-1} = (x^{-(a+b)+a})^{-1} = (x^{-(a+b)} x^a)^{-1} = (x^a)^{-1}(x^{-(a+b)})^{-1} = x^{-a} x^{a+b}$. Then $x^a x^b = x^{a+b}$.

For the second part of (a), we have $(x^a)^b = ((x^a)^{-b})^{-1} = (x^{-ab})^{-1} = x^{ab}$. Next, the case of $a < 0$ and $b > 0$ is similar. Lastly, if $a < 0$ and $b < 0$, then $x^{a+b} = (x^{-b-a})^{-1} = (x^{-b} x^{-a})^{-1} = (x^{-a})^{-1}(x^{-b})^{-1} = x^a x^b$. Additionally, $(x^a)^b = ((x^a)^{-b})^{-1} = (x^{-ab})^{-1} = ((x^{ab})^{-1})^{-1} = x^{ab}$. □

(20) For x an element in G, show that x and x^{-1} have the same order.

Solution. Let $|x| = n \in \mathbb{Z}^+$. Then $(x^{-1})^n = (x^n)^{-1} = 1^{-1} = 1$, so $|x^{-1}| \le |x|$. If $|x^{-1}| = m \in \mathbb{Z}^+$, then $x^m = (x^{-1})^{-m} = ((x^{-1})^m)^{-1} = 1^{-1} = 1$, ,so $|x| \le |x^{-1}|$. Hence, $n = m$. This establishes that the orders are the same if it is finite; otherwise, both $|x|$ and $|x^{-1}|$ must be of infinite order. □

(21) Let G be a finite group and let x be an element of order n. Prove that if n is odd, then $x = (x^2)^k$ for some $k \ge 1$.

Solution. Put $n = 2k - 1$ for $k \ge 1$. Then $1 = x^n = x^{2k-1} = (x^2)^k x^{-1}$. The result follows. □

(22) If x and g are elements of the group G, prove that $|x| = |g^{-1}xg|$. Deduce that $|ab| = |ba|$ for all $a, b \in G$.

Solution. We first prove that $(g^{-1}xg)^n = g^{-1}x^n g$ by induction. For $n = 1$, the result is clear. Supposing it is true for some $n = k$, then for $n = k+1$, we have $(g^{-1}xg)^{k+1} = (g^{-1}xg)^k(g^{-1}xg) = (g^{-1}x^k g)(g^{-1}xg) = g^{-1}x^{k+1}g$.

Put $|x| = n \in \mathbb{Z}^+$. Then $(g^{-1}xg)^n = g^{-1}x^n g = g^{-1}1g = 1$ so that $|g^{-1}xg| \le n$. Similarly, put $|g^{-1}xg| = m$. Then $1 = (g^{-1}xg)^m = g^{-1}x^m g$. Then $1 = x^m$ so that $|x| \le m$, hence $|x| = |g^{-1}xg|$. Similar to the previous exercise, then both must be of finite or infinite order. Finally, for $a, b \in G$, we have $|ab| = |b(ab)^{-1}| = |ba|$. □

(23) Suppose $x \in G$ and $|x| = n < \infty$. If $n = st$ for some positive integers s and t, prove that $|x^s| = t$.

Solution. Let $|x^s| = r$. Since $1 = x^n = x^{st} = (x^s)^t$, then $|x^s| \le t$. Moreover, we have $(x^s)^r = x^{sr} = 1$ so that $|x| = st \le sr$. Then $t \le |x^s|$. □

(24) If a and b are commuting elements of G, prove that $(ab)^n = a^n b^n$ for all $n \in \mathbb{Z}$. [Do this by induction for positive n first.]

Solution. The case for $n = 1$ is clear. Supposing the result holds for $n = k$, then for $n = k+1$, we have $(ab)^{k+1} = (ab)^k(ab) = a^k b^k ab = a^{k+1}b^{k+1}$. The result for $n = 0$ is trivial, and for $n < 0$, we have $(ab)^n = (ba)^n = ((ba)^{-n})^{-1} = (b^{-n}a^{-n})^{-1} = a^n b^n$. □

(25) Prove that if $x^2 = 1$ for all $x \in G$, then G is abelian.

Solution. Note that $x^2 = 1$ implies $x = x^{-1}$ for any $x \in G$. Then for $a, b \in G$, we have that $ab = (ab)^{-1} = b^{-1}a^{-1} = ba$. □

(26) Assume H is a nonempty subset of $(G, \star)$ which is closed under the binary operation on G and is closed under inverses, i.e., for all $h, k \in H$, hk, $h^{-1} \in H$. Prove that H is a group under the operation $\star$ restricted to H (such a subset H is called a *subgroup* of G).

Solution. Since $hk, h^{-1} \in H$, then H has existence of inverse and closure. Moreover, $\star$ is associative in H due to associativity in G. Lastly, there exists $h \in H$ since it is nonempty. Then $h^{-1} \in H$ so that $hh^{-1} = 1 \in H$. Hence, $(H, \star)$ is a group. □

(27) Prove that if x is an element of the group G, then $\{x^n : n \in \mathbb{Z}\}$ is a subgroup. (cf. the preceding exercise) of G (called the *cyclic subgroup* of G generated by x).

Solution. Let $x^m, x^n \in G$. Then $x^m x^n = x^{m+n} \in G$ since $m+n \in \mathbb{Z}$. Moreover, $x^{-n} \in G$ since $-n \in \mathbb{Z}$ so that the set has an inverse. Additionally, $x^0 = 1$ is in the set. Then the set is a subgroup of G. □

(28) Let $(A, \star)$ and $(B, \diamond)$ be groups and let $A \times B$ be their direct product (as defined in Example 6). Verify all the group axioms for $A \times B$:

(a) Prove that the associative law holds:

$$\text{for all } (a_1, b_1), (a_2, b_2), (a_3, b_3) \in A \times B, [(a_1, b_1)(a_2, b_2)](a_3, b_3) = (a_1, b_1)[(a_2, b_2)(a_3, b_3)]$$

(a) Prove that $(1, 1)$ is the identity of $A \times B$, and

(b) Prove that the inverse of (a, b) is (a^{-1}, b^{-1}).

Solution.

(a) Let $(a_1, b_1), (a_2, b_2), (a_3, b_3) \in A \times B$. Then

$$\begin{aligned}
[(a_1, b_1)(a_2, b_2)](a_3, b_3) &= (a_1a_2, b_1b_2)(a_3, b_3) \\
&= ((a_1a_2)a_3, (b_1b_2)b_3) \\
&= (a_1(a_2a_3), b_1(b_2b_3)) \\
&= (a_1, b_1)(a_2a_3, b_2b_3) \\
&= (a_1, b_1)[(a_2, b_2)(a_3, b_3)]
\end{aligned}$$

(b) For $a \in A, b \in B$, we have $(a, b)(1, 1) = (a \star 1, b \diamond 1) = a, b)$.

(c) We have $(a, b)(a^{-1}, b^{-1}) = (a \star a^{-1}, b \diamond b^{-1}) = (1, 1)$. □

(29) Prove that $A \times B$ is an abelian group if and only if both A and B are abelian.

Solution. Let $a, a' \in A$ and $b, b' \in B$. Suppose $A \times B$ is abelian. Then

$$(aa', bb') = (a, b)(a', b') = (a', b')(a, b) = (a'a, b'b)$$

so that A and B are abelian. If A and B are both abelian, then

$$(a, b)(a', b) = (aa', bb') = (a'a, b'b) = (a', b')(a, b)$$

hence $A \times B$ is abelian. □

(30) Prove that the elements $(a, 1)$ and $(1, b)$ of $A \times B$ commute and deduce that the order of (a, b) is the least common multiple of $|a|$ and $|b|$.

Solution. We have $(a, 1)(1, b) = (a1, 1b) = (1a, b1) = (1, b)(a, 1)$. Let $\ell = \text{lcm}(|a|, |b|)$ so that $\ell = m|a| = n|b|$ for $m, n \in \mathbb{Z}$. Put $|(a, b)| = k$. Since $(a, b)^k = (a^k, b^k) = (1, 1)$, then $|a|$ and $|b|$ must divide k so that $\ell \le k$. Moreover, we have $(a, b)^\ell = (a^\ell, b^\ell) = (a^{m|a|}, b^{n|b|}) = (1^m, 1^n) = (1, 1)$ so that $k \le \ell$. The result follows. □

(31) Prove that any finite group G of even order contains an element of order 2. [Let $t(G)$ be the set $\{g \in G \mid g \neq g^{-1}\}$.] Show that $t(G)$ has an even number of elements and every nonidentity element of $G - t(G)$ has order 2.]

Solution. Note that if $g \in G$, then $g^{-1} \in G$ so that the elements of $t(G)$ come in pairs. Then $|t(G)|$ is even, and $|G - t(G)|$ is also even. Moreover, $1 \notin t(G)$ since $1^{-1} = 1$. Then $1 \in G - t(G)$ so that it is nonempty, and there must exist $h \in G - t(G)$ such that $h = h^{-1}$ and $h \neq 1$ (because $|G - t(G)| \le 2$). Then $h^2 = 1$. □

(32) If x is an element of finite order n in G, prove that the elements $1, x, x^2, \ldots, x^{n-1}$ are all distinct. Deduce that $|x| \le |G|$.

Solution. Suppose x^a, x^b are not distinct for $1 \le a < b < n$. Then $x^{b-a} = 1$. But then $b-a < n$, contradicting that $|x| = n$. It follows that $|x| \le |G|$. $\square$

(33) Let x be an element of finite order in G.

(a) Prove that if n is odd then $x^i \neq x^{-i}$ for all $i = 1, 2, \dots, n-1$.

(b) Prove that if $n = 2k$ and $1 \le i < n$ then $x^i = x^{-i}$ if and only if $i = k$.

Solution.

(a) Let $i \le n$. Then $x^i x^{n-i} = 1$ and $x^{-i} = x^{n-1}$. By Exercise (32), then $x^i \neq x^{n-i}$ when $i \neq n-i$. Then $x^i \neq x^{-i}$. For odd n, certainly $i \neq i$ so the result follows.

(b) For $1 \le i < n$ where $i \neq k$, then $i \neq n-i$ so that $x^i \neq x^{n-i}$ by the previous part. Moreover, if $i = k$, then $x^i x^i = 1$ and $x^i = x^{-i}$. $\square$

(34) If x is an element of infinite order in G, prove that the elements $x^n, n \in \mathbb{Z}$ are all distinct.

Solution. Suppose $x^m = x^n$ for $m \neq n$. Then $x^{m-n} = 1$, which is a contradiction. $\square$

(35) If x is an element of finite order in G, use the Division Algorithm to show that any integral power of x equals one of the elements in the set $\{1, x, x^2, \dots, x^{n-1}\}$ (so these are all the distinct elements of the cyclic subgroup (cf. Exercise 27 above) of G generated by x).

Solution. Let $x^k \in G$ and $|x| = n$. Then $k = nq + r$ for $0 \le r < n$ and $q \in \mathbb{Z}$. Then $x^k = x^{nq+r} = x^{nq} x^r = 1x^r = x^r$. $\square$

(36) Assume $G = \{1, a, b, c\}$ is a group of order 4 with identity 1. Assume also that G has no elements of order 4 (so by Exercise 32, every element has order ≤ 3). Use the cancellation laws to show that there is a unique group table for G. Deduce that G is abelian.

Solution. Since no element has order 4, then each of the elements must have order 2. Now consider ab (without loss of generality; we may use bc or ac). If $ab = 1$, then $a = b^{-1}$, which cannot be true since $b = b^{-1}$ because $b^2 = 1$. Moreover, $ab \neq a$ since $b \neq 1$. Similarly, $ab \neq b$. It must be that $ab = c$. Moreover, $a^2 b = ac$ so that $ba^{-1} = aca^{-1} = a(ab)a = ba$. Using similar reasoning, we deduce that $bc = cb = a$ and $ac = ca = b$. Then we have the group table:

$\cdot$	1	a	b	c
1	1	a	b	c
a	a	1	c	b
b	b	c	1	a
c	c	b	a	1

$\square$

1.2 Dihedral Groups

In these exercises, D_{2n} has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$.

(1) Compute the order of each of the elements in the following groups:

(a) D_6

(b) D_8

(c) D_{10}

Solution. Note that for any sr^k for $1 \le k \le n-1$, we have $(sr^k)(sr^k) = s(r^k s)r^k = s(sr^{-k})r^k = s^2 = 1$ so that $|sr^k| = 2$. All that's left for the rest of the exercises is to calculate the order of the rotations:

(a) $D_6 = \{1, r, r^2, s, sr, sr^2\}$. Then $|1| = 1$ and $|r| = |r^2| = 2$.

(b) $D_8 = \{1, r, r^2, r^3, s, sr, sr^2, sr^3\}$. Then $|1| = 1, |r^2| = 2$ and $|r| = |r^3| = 4$.

(c) $D_{10} = \{1, r, r^2, r^3, r^4, s, sr, sr^2, sr^3, sr^4\}$. Then $|1| = 1$ and $|r| = |r^2| = |r^3| = |r^4| = 5$. $\square$

(2) Use the generators and relations above to show that if x is any element of D_{2n} which is not a power of r, then $rx = xr^{-1}$.

***Solution*.** If $x \neq r^k$, then $x = sr^k$. Then $r(sr^k) = (sr^{-1})r^k = sr^k r^{-1}$. □

(3) Use the generators and relations above to show that every element of D_{2n} which is not a power of r has order 2. Deduce that D_{2n} is generated by the two elements s and sr, both of which have order 2. [cf. Exercise 33 of Section 1.]

***Solution*.** See Exercise (1) from this section for the solution to the first part. For any $r^i \in D_{2n}$, then $(s(sr))^i = r^i$, and for any $sr^i \in D_{2n}$, then $s(s(sr))^i = sr^i$ so that D_{2n} is generated by $\{s, sr\}$. □

(4) If $n = 2k$ is even and $n \geq 4$, show that $z = r^k$ is an element of order 2 which commutes with all elements of D_{2n}. Show also that z is the only nonidentity element of D_{2n} which commutes with all elements of D_{2n}. [cf. Exercise 33 of Section 1.]

***Solution*.** Clearly $r^k \neq 1$, and $(r^k)^2 = r^n = 1$ so that $|r^k| = 2$. Noting that $r^k = r^{-k}$ since $r^{2k} = 1$, then for any $s^i r^j$ where $i \in \{0,1\}$ and $1 \leq j \leq n-1$, then $(s^i r^j)r^k = (s^i r^k)r^j = (r^{-k}s^i)r^j = r^k(s^i r^j)$ so that r^k commutes with any element.

Suppose now that there existed a nonidentity element $w \in D_{2n}$ that commutes with every element of D_{2n}. In particular, w must commute with s, i.e., $sw = ws$. There are two cases to discuss:

- If $w = r^t$ for some $t \in \mathbb{Z}$, then $sw = sr^t$ and $ws = r^t s$. Then $sr^t = r^t s$, or $sr^t = sr^{-t}$, implying that $t = k$ by Section 1, Exercise (33).
- If $w = sr^t$, then $sw = r^t$ and $ws = r^{-t}$, implying that $t = 0$ or $t = k$. Then $w = s$ or $w = sr^k$. Since s does not commute with r, then that cannot be. If $w = sr^k$, then $sr^{k+1} = rsr^k = sr^{k-1}$, or that $k = 2$, contradicting that $n \geq 4$. Hence, it follows that w cannot be a reflection.

Then $w = r^k = z$, making it unique. □

(5) If n is odd and $n \geq 3$, show that the identity is the only element of D_{2n} which commutes with all elements of D_{2n}. [cf. Exercise 33 of Section 1.]

***Solution*.** Use the odd case in Section 1, Exercise (33). □

(6) Let x and y be elements of order 2 in any group G. Prove that if $t = xy$, then $tx = xt^{-1}$ (so that if $n = |xy| < \infty$ then x, y satisfy the same relations in G as s, r do in D_{2n}).

***Solution*.** Note that $|x| = |y| = 2$, so $x = x^{-1}$ and $y = y^{-1}$. Then $tx = xyx = xy^{-1}x^{-1} = xt^{-1}$. □

(7) Show that $\langle a, b \mid a^2 = b^2 = (ab)^n = 1 \rangle$ gives a presentation for D_{2n} in terms of the two generators a and b of order 2 computed in Exercise 3 above. [Show that the relations for r and s follow from the relations for a and b and conversely, the relations for a and b follow from those for r and s.]

***Solution*.** Suppose $a^2 = b^2 = (ab)^n = 1$. The first natural choice is to let $a = s$ so that $s^2 = 1$. The next one is that $ab = r$ so that $r^n = 1$. Since $sb = r$, then $b = s^{-1}r = sr$. Moreover, $sr = b = b^{-1} = r^{-1}s^{-1} = r^{-1}s$ so that we have the regular presentation of D_{2n}.

Using the regular representation, then $a^2 = s^2 = 1$ and $b^2 = srsr = ssr^{-1}r = 1$. Moreover, we have $(ab)^n = (s(sr))^n = r^n = 1$, and we have the resulting presentation. □

(8) Find the order of the cyclic subgroup of D_{2n} generated by r (cf. Exercise 27 of Section I).

***Solution*.** Let $G = \langle r \rangle$ be the cyclic subgroup of D_{2n} generated by r. Clearly $\{1, r, r^2, \ldots, r^{n-1}\} \subseteq G$ so that $|G| \geq n$. Moreover, for $r^k \in G$, we may use the division algorithm to write $k = nq + x$ for $0 \leq r < n$, and $r^k = r^{nq+x}$ so that $r^k = r^x$, where r^x is any one of the elements in G by definition. Then $G \subseteq \{1, r, r^2, \ldots, r^{n-1}\}$ so that $|G| = n$. □

In each of Exercises 9 to 13 you can find the order of the group of rigid motions in $\mathbb{R}^3$ (also called the group of rotations) of the given Platonic solid by following the proof for the order of D_{2n}: find the number of positions to which an adjacent pair of vertices can be sent. Alternatively, you can find the number of places to which a given face may be sent and, once a face is fixed, the number of positions to which a vertex on that face may be sent.

(9) Let G be the group of rigid motions in $\mathbb{R}^3$ of a tetrahedron. Show that $|G| = 12$.

Solution. Label the vertices of a tetrahedron 1 through 4 so that vertex 1 has 4 choices. Then vertex 2 has 3 remaining choices, and vertex 3 and 4 are determined after. It follows that $4(3) = 12$ symmetries. $\square$

(10) Let G be the group of rigid motions in $\mathbb{R}^3$ of a cube. Show that $|G| = 24$.

Solution. 8 choices for vertex 1 and 3 adjacent vertices for vertex 2; $8(3) = 24$ symmetries. $\square$

(11) Let G be the group of rigid motions in $\mathbb{R}^3$ of an octahedron. Show that $|G| = 24$.

Solution. 6 choices for vertex 1 and 4 adjacent vertices for vertex 2; $6(4) = 24$ symmetries. $\square$

(12) Let G be the group of rigid motions in $\mathbb{R}^3$ of a dodecahedron. Show that $|G| = 60$.

Solution. 20 choices for vertex 1 and 3 adjacent vertices for vertex 2; $20(3) = 60$ symmetries. $\square$

(13) Let G be the group of rigid motions in $\mathbb{R}^3$ of an icosahedron. Show that $|G| = 60$.

Solution. 12 choices for vertex 1 and 5 adjacent vertices for vertex 2; $12(5) = 60$ symmetries. $\square$

(14) Find a set of generators for $\mathbb{Z}$.

Solution. Any integer is of the form $n1$, so $\mathbb{Z} = \langle 1 \rangle$. $\square$

(15) Find a set of generators and relations for $\mathbb{Z}/n\mathbb{Z}$.

Solution. Each element of $\mathbb{Z}/n\mathbb{Z}$ is of the form $k\bar{1}$ for $0 \le k < n$. Then $\mathbb{Z}/n\mathbb{Z} = \langle \bar{1} \mid n\bar{1} = \bar{0} \rangle$. $\square$

(16) Show that the group $\langle x_1, y_1 \mid x_1^2 = y_1^2 = (x_1 y_1)^2 = 1 \rangle$ is the dihedral group D_4 (where x_1 may be replaced by the letter r and y_1 by s). (Show that the last relation is the same as $x_1 y_1 = y_1 x_1^{-1}$.)

Solution. Note that $D_4 = \langle r, s \mid r^2 = s^2 = 1, rs = sr^{-1} \rangle$. Note that $r = r^{-1}$ so that $rs = sr$. Put $x_1 = r$ and $y_1 = s$. If $rs = sr$, then $(x_1 y_1)^2 = rsrs = r^2 s^2 = 1$. Moreover, if $(x_1 y)^2 = 1$, then $rsrs = 1$ so that $rs = sr$. $\square$

(17) Let X_{2n} be the group whose presentation is displayed in (1.2).

(a) Show that if $n = 3k$, then X_{2n} has order 6, and it has the same generators and relations as D_6 when x is replaced by r and y by s.

(b) Show that if $(3, n) = 1$, then x satisfies the additional relation: $x = 1$. In this case deduce that X_{2n} has order 2. [use the facts that $x^n = 1$ and $x^3 = 1$.]

Solution.

(a) Applying $n = 3k$, we have $X_{2n} = X_{6k} = \langle x, y \mid x^{3k} = y^2 = 1, xy = yx^2 \rangle$. Now suppose the given relations, and recall the text showing $x^3 = 1$. Set $x = r$ and $y = s$. Then $r^3 = 1$ and $s^2 = 1$. From $xy = yx^2$, we have $rs = sr^2 = sr^{-1}$.

Suppose now that $r^3 = s^2 = 1$ and $rs = sr^{-1}$. Then $x^{3k} = (x^3)^k = 1$. Moreover, $s^2 = y^2 = 1$, and $xy = rs = sr^{-1} = sr^2 = yx^2$. Then the two representations are equal.

(b) Note that $(3, n) = 1$ implies that $3a + nb = 1$ for some $a, b \in \mathbb{Z}$. Then $x = x^{3a+nb} = x^{3a} x^{nb} = (x^3)^a (x^n)^b = 1$. Then $X_{2n} = \{1, y\}$, and $|X_{2n}| = 2$. $\square$

(18) Let Y be the group whose presentation is displayed in (1.3).

(a) Show that $v^2 = v^{-1}$. [Use the relation $v^3 = 1$.]

(b) Show that v commutes with u^3. [Show that $v^2 u^3 v = u^3$ by writing the left-hand side as $(v^2 u^2)(uv)$ and using the relations to reduce this to the right-hand side. Then use part (a).]

(c) Show that v commutes with u. [Show that $u^9 = u$ and then use part (b).]

(d) Show that $uv = 1$. [Use part (c) and the last relation.]

(e) Show that $u = 1$. Deduce that $v = 1$, and conclude that $Y = 1$. [Use part (d) and the equation $u^4 v^3 = 1$.]

Solution.

(a) $v^3 = 1 \implies v^2 = v^{-1}$.
(b) From $uv = v^2u^2$, we have $vuv = u^2$ so that $vu = u^2v^{-1} = u^2v^2$. It follows that $v^2u^3v = (v^2u^2)(uv) = (uv)(uv) = u(vu)v = u(u^2v^2)v = u^3$.
(c) It follows that $u^9 = (u^4)^2u = u$. Then $uv = (u^3)^3v = v(u^3)^3 = vu$.
(d) $uv = v^2u^2 = u^2v^2$. Then $1 = uv$.
(e) $u^4v^3 = u(uv)^3 = u1^3 = 1$. Then $1v = 1$, and $u = v = 1$, so $Y = 1$. □

1.3 Symmetric Groups

(1) Let σ be the permutation

$$1 \mapsto 3, \quad 2 \mapsto 4, \quad 3 \mapsto 5, \quad 4 \mapsto 2, \quad 5 \mapsto 1$$

and let τ be the permutation

$$1 \mapsto 5, \quad 2 \mapsto 3, \quad 3 \mapsto 2, \quad 4 \mapsto 4, \quad 5 \mapsto 1.$$

Find the cycle decompositions of each of the following permutations: $\sigma, \tau, \sigma^2, \sigma\tau, \tau\sigma$, and $\tau^2\sigma$.

Solution.

$$\begin{aligned}
\sigma &= (1\ 3\ 5)(2\ 4) \\
\tau &= (1\ 5)(2\ 3) \\
\sigma^2 &= (1\ 5\ 3) \\
\sigma\tau &= (2\ 5\ 3\ 4) \\
\tau\sigma &= (1\ 2\ 4\ 3) \\
\tau^2\sigma &= (1\ 3\ 5)(2\ 4)
\end{aligned}$$

□

(2) Let σ be the permutation

$$\begin{array}{ccccc}
1 \mapsto 13 & 2 \mapsto 2 & 3 \mapsto 15 & 4 \mapsto 14 & 5 \mapsto 10 \\
6 \mapsto 7 & 7 \mapsto 12 & 8 \mapsto 9 & 9 \mapsto 3 & 10 \mapsto 1 \\
11 \mapsto 7 & 12 \mapsto 9 & 13 \mapsto 5 & 14 \mapsto 11 & 15 \mapsto 8
\end{array}$$

and let τ be the permutation

$$\begin{array}{ccccc}
1 \mapsto 14 & 2 \mapsto 9 & 3 \mapsto 10 & 4 \mapsto 2 & 5 \mapsto 12 \\
6 \mapsto 6 & 7 \mapsto 5 & 8 \mapsto 11 & 9 \mapsto 15 & 10 \mapsto 3 \\
11 \mapsto 8 & 12 \mapsto 7 & 13 \mapsto 4 & 14 \mapsto 1 & 15 \mapsto 13
\end{array}$$

Find the cycle decompositions of the following permutations: $\sigma, \tau, \sigma^2, \sigma\tau, \tau\sigma$, and $\tau^2\sigma$.

Solution.

$$\begin{aligned}
\sigma &= (1\ 13\ 5\ 1)(3\ 15\ 8)(4\ 14\ 11\ 7\ 12\ 9) \\
\tau &= (1\ 14)(2\ 9\ 15\ 13\ 4)(3\ 10)(5\ 12\ 7)(8\ 11) \\
\sigma^2 &= (1\ 5)(3\ 8\ 15)(4\ 11\ 12)(7\ 9\ 14)(10\ 13) \\
\sigma\tau &= (1\ 11\ 3)(2\ 4)(5\ 9\ 8\ 7\ 10\ 15)(8\ 10\ 14) \\
\tau\sigma &= (1\ 4)(2\ 9)(3\ 13\ 12\ 15\ 11\ 5)(8\ 10\ 14) \\
\tau^2\sigma &= (1\ 2\ 15\ 8\ 3\ 4\ 14\ 11\ 12\ 13\ 7\ 5\ 10)
\end{aligned}$$

□

(3) For each of the permutations whose cycle decompositions were computed in the preceding two exercises, compute its order.

Solution. Note that the order of a cycle decomposition is the lcm of the orders of each of the cycles, and that the order of an m-cycle is m. Then the orders are the following (the first number will be for Exercise 1, and the second for Exercise 2):

$$\begin{aligned}|\sigma| &= 6, 12\\ |\tau| &= 2, 30\\ |\sigma^2| &= 3, 6\\ |\sigma\tau| &= 4, 6\\ |\tau\sigma| &= 4, 6\\ |\tau^2\sigma| &= 6, 13\end{aligned}$$

□

(4) Compute the order of each of the elements in the following groups:
   (a) S_3
   (b) S_4

Solution.

(a)

Permutation	Order
1	1
(12)	2
(13)	2
(23)	2
(123)	3
(132)	3

(b)

Permutation	Order	Permutation	Order	Permutation	Order	Permutation	Order
1	1	(34)	2	(143)	3	(1342)	4
(12)	2	(123)	3	(234)	3	(1423)	4
(13)	2	(124)	3	(243)	3	(1432)	4
(14)	2	(132)	3	(1234)	4	(12)(34)	2
(23)	2	(134)	3	(1243)	4	(13)(24)	2
(24)	2	(142)	3	(1324)	4	(14)(23)	2

□

(5) Find the order of (1 12 8 10 4)(2 13)(5 11 7)(6 9).

Solution. $\mathrm{lcm}(5, 2, 3) = 30$. □

(6) Write out the cycle decomposition of each element of order 4 in S_4.

Solution. See Exercise 4. □

(7) Write out the cycle decomposition of each element of order 2 in S_4.

Solution. See Exercise 4. □

(8) Prove that if $\Omega = \{1, 2, 3, \dots\}$, then S_Ω is an infinite group (do not say $\infty! = \infty$).

Solution. Consider the permutation $\sigma_n = (2n-1\ 2n)$. Note that $\sigma_i \neq \sigma_j$ for any $i, j \in \mathbb{Z}^+$ so that the set $A = \{\sigma_n \mid n \in \mathbb{Z}^+\}$ are all distinct. Moreover, A is an infinite subset of S_Ω so that $|S_\Omega| = \infty$. □

(9) (a) Let σ be the 12-cycle (1 2 3 4 5 6 7 8 9 10 11 12). For which positive integers i is σ^i also a 12-cycle?
   (b) Let τ be the 8-cycle (1 2 3 4 5 6 7 8). For which positive integers i is τ^i also an 8-cycle?

(c) Let ω be the 14-cycle (1 2 3 4 5 6 7 8 9 10 11 12 13 14). For which positive integers i is ω^i also a 14-cycle?

Solution.

(a) We will compute the explicit powers for the first problem, notice a pattern with the integers that produce a 12-cycle, and apply that pattern to the other problems.

For each integer between 1 and 11, we compute the cycle:

$$\begin{aligned}
\sigma^1 &= \sigma \\
\sigma^2 &= (1\ 3\ 5\ 7\ 9\ 11)(2\ 4\ 6\ 8\ 10\ 12) \\
\sigma^3 &= (1\ 4\ 7\ 10)(2\ 5\ 8\ 11)(3\ 6\ 9\ 12) \\
\sigma^4 &= (1\ 5\ 9)(2\ 6\ 10)(3\ 7\ 11)(4\ 8\ 12) \\
\sigma^5 &= \sigma \\
\sigma^6 &= (1\ 7)(2\ 8)(3\ 9)(4\ 10)(5\ 11)(6\ 12) \\
\sigma^7 &= \sigma \\
\sigma^8 &= \sigma^4 \\
\sigma^9 &= \sigma^3 \\
\sigma^{10} &= \sigma^2 \\
\sigma^{11} &= \sigma
\end{aligned}$$

From direct calculations, it seems that the integers i such that $(12, i) = 1$ produce a 12-cycle, while i such that $(12, i) = k$ produces k $12/k$-cycles. In this particular case, the set of integers that produce a 12-cycle is the set $\{x + 12k \mid x \in \{1, 5, 7, 11\}, k \in \mathbb{Z}\}$.

(b) $\{x + 8k \mid x \in \{1, 3, 5, 7\}, k \in \mathbb{Z}\}$.

(c) $\{x + 14k \mid x \in \{1, 3, 5, 9, 11, 13\}, k \in \mathbb{Z}\}$. □

(10) Prove that if σ is the m-cycle $(a_1\ a_2\ \dots\ a_m)$, then for all $i \in \{1, 2, \dots, m\}$, $\sigma^i(a_k) = a_{k+i}$, where $k+i$ is replaced by its least residue mod m when $k + i > m$. Deduce that $|\sigma| = m$.

Solution. We prove the first part by induction. Clearly for $i = 1$, this holds true by definition of cycles. Suppose it holds true for some $i \in \mathbb{Z}^+$. Then $\sigma^{i+1}(a_k) = \sigma(\sigma^i(a_k)) = \sigma(a_{k+i}) = a_{k+i+1}$.

Now suppose $1 \le i < m$. Then $\sigma^i(a_1) = a_{1+i} \neq a_1$. But then $\sigma^m(a_1) = a_{1+m} = a_1$ since $(1+m) \mod m = 1$. Then $|\sigma| = m$. □

(11) Let σ be the m-cycle $(1\ 2\ \dots\ m)$. Show that σ^i is also an m-cycle if and only if i is relatively prime to m.

Solution. Suppose σ^i is an m-cycle, and put $(m, i) = d$. Then there exists $a, b \in \mathbb{Z}^+$ such that $ad = i$ and $bd = m$. Noting that $|\sigma^i| = m$ because it is an m-cycle, then $(\sigma^i)^b = (\sigma^{ad})^b = (\sigma^{bd})^a = (\sigma^m)^a = 1^a = 1$. Then $|\sigma^i| \le b$, or $m = b$. It follows that $d = 1$.

Suppose that $(m, i) = 1$, and denote $x' \equiv x \mod m$. Note that the integers $(1+i)', (1+2i)', \dots, (1+(m-1)i)'$ are all distinct, since for any $0 \le x, y \le m-1$ such that $x \neq y$, then $xi \equiv yi \mod m$ implies $x = y$ (This is seen as follows: $i(x-y) \equiv 0 \mod m$ implies that $m \mid (x-y)$, since $(m, i) = 1$. Since $1 - m \le x - y \le m - 1$, then the only choice for $x - y$ is 0, or that $x = y$). Then we have m integers so that

$$\sigma^i = (1\ (1+i)'\ (1+2i)' \dots (1+(m-1)i)'),$$

and σ^i is an m-cycle. □

(12) (a) If $\tau = (1\ 2)(3\ 4)(5\ 6)(7\ 8)(9\ 10)$, determine whether there is an n-cycle ($n \ge 10$) with $\tau = \sigma^k$ for some integer k.

(b) If $\tau = (1\ 2)(3\ 4\ 5)$, determine whether there is an n-cycle ($n \ge 5$) with $\tau = \sigma^k$ for some integer k.

Solution.

(a) $\sigma = (1\ 3\ 5\ 7\ 9\ 2\ 4\ 6\ 8\ 10), \sigma^5 = \tau$.

(b) We first prove that $(ac, bc) = c(a, b)$. Let $d = (a, b)$ and $d' = (ac, bc)$. Then there exist $x, y \in \mathbb{Z}$ such that $ax + by = d$. Then $acx + bcy = dc$. Since d' is the smallest integer that is written of the form $acx' + bcy'$ for $x', y' \in \mathbb{Z}$ (if there were any greater, then $d' \neq (ac, bc)$), then $d' \leq dc$. Moreover, $d \mid a$ and $d \mid b$ implies $cd \mid ac$ and $cd \mid bc$. Then $cd \mid d'$, or $cd \leq d'$. Then $dc = d'$, or $c(a, b) = (ac, dc)$.

Suppose there existed some n-cycle σ and k such that $\sigma^k = \tau$. Then $\sigma^{2k}(1) = 1$, or $2k \equiv 0 \bmod n$. Moreover, $\sigma^{3k}(3) = 3$, or that $3k \equiv 0 \bmod n$. Then $n \mid 2k$ and $n \mid 3k$ so that $n \mid (2k, 3k)$. By the above, then $(2k, 3k) = k(2, 3) = k$, and $n \mid k$. But then $\sigma^k = (\sigma^n)^q = 1$, contradicting that $\sigma^k = \tau$. □

(13) Show that an element has order 2 in S_n if and only if its cycle decomposition is a product of commuting 2-cycles.

Solution. Suppose $\sigma \in S_n$ such that $|\sigma| = 2$. Perform cycle decomposition on σ so that it becomes a product of commuting, disjoint cycles. Pick one of the cycles $(a_1\ a_2 \dots a_m)$. Since $\sigma \neq 1$, then $\sigma(a_1) = a_2$ and $\sigma^2(a_1) = a_3$. However, $\sigma^2 = 1$, so $\sigma^2(a_1) = a_1$ so that $a_1 = a_3$. Then $m \leq 2$, and the cycle is $(a_1\ a_2)$ so that any cycle in the cycle decomposition of σ is of length 2.

Suppose $\sigma = (a_1\ b_1)(a_2\ b_2)\dots(a_k\ b_k)$. Since $(a_1\ b_1)^2 = 1$, and the 2-cycles commute, then $\sigma^2 = 1$. Moreover, because $\sigma \neq 1$, then $|\sigma| = 2$. □

(14) Let p be a prime. Show that an element has order p in S_n if and only if its cycle decomposition is a product of commuting p-cycles. Show by an explicit example that this need not be the case if p is not prime.

Solution. Suppose $\sigma \in S_n$ such that $|\sigma| = p$ that is prime. Put

$$\sigma = \pi_1 \pi_2 \dots \pi_k$$

as the cycle decomposition of σ so that π_i commutes with π_j for any $i \neq j$. Clearly, $\sigma^p = \pi_1^p \pi_2^p \dots \pi_k^p$. Since $\sigma^p = 1$, then each $\pi_i^p = 1$. Then π_i must be a p-cycle or $\pi_i = 1$. But no $\pi_i = 1$, since $\sigma \neq 1$. Then π_i must be a p-cycle.

Suppose σ's cycle decomposition is a product of commuting p-cycles π_i. Then for any $t < p$, we have

$$\sigma^t = \pi_1^t \pi_2^t \dots \pi_k^t \neq 1$$

since π_i is a p-cycle. If $t = p$, then $\sigma^t = 1$. Then $|\sigma| = p$. □

(15) Prove that the order of an element in S_n equals the least common multiple of the lengths of the cycles in its cycle decomposition.

Solution. Let $\sigma \in S_n$ have the cycle decomposition $\sigma = \pi_1 \pi_2 \dots \pi_k$, and let π_i have length p_i so that $|\pi_i| = p_i$. Suppose $|\sigma| = m$. Then $\pi_i^m = 1$ so that $p_i \mid m$. In particular, every p_i must divide m so that m is a common multiple of all p_i. Moreover, if there was another t such that $\sigma^t = 1$, then t must be a common multiple of all p_i and $m \leq t$ so that m is the least common multiple of all p_i. □

(16) Show that if $n \geq m$ then the number of m-cycles in S_n is given by

$$\frac{n(n-1)(n-2)\dots(n-m+1)}{m}.$$

[Count the number of ways of forming an m-cycle and divide by the number of representations of a particular m-cycle.]

Solution. Note that in an m-cycle, there are m integers to write. The first integer has n choices, the second $n-1$ choices. Continuing on, then the m-th integer has $n-m+1$ choices. However, each choice of m integers has m equivalent representations (such as (1 2 3 4) being equivalent to (2 3 4 1) by shifting each integer to the left one place). Then take the product $n(n-1)(n-2)\dots(n-m+1)$ and divide by m. □

(17) Show that if $n \geq 4$ then the number of permutations in S_n which are the product of two disjoint 2-cycles is $n(n-1)(n-2)(n-3)/8$.

Solution. The first 2-cycle has $n(n-1)/2$ choices, and the second 2-cycle has $(n-2)(n-3)/2$ choices. Moreover, there are 2 representations for the same set of 2-cycles, so multiply the above quantities and divide by 2 to obtain $n(n-1)(n-2)(n-3)/8$. □

(18) Find all numbers n such that S_5 contains an element of order n.

Solution. S_5 contains elements of order 1 through 5. Moreover, we can have a 2 and 3-cycle together whose lcm is 6. Then $n = 1, 2, 3, 4, 5, 6$. $\square$

(19) Find all numbers n such that S_7 contains a number an element of order n.

Solution. We have n from 1 to 7. We can have 2 and 5-cycles and 3 and 4-cycles. Then $n = 10$ and 12. $\square$

(20) Find a set of generators and relations for S_3.

Solution. Recall that $S_3 = \{1, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$. Moreover, no element in S_3 has order 6 so that no element generates S_3. It must be that there are at least 2 generators. Put $\alpha = (1\ 2)$ and $\beta = (1\ 3)$. Then $\alpha^2 = \beta^2 = 1$. Then $|\alpha| = |\beta| = 2$, and $\alpha\beta = (1\ 3\ 2), \beta\alpha = (1\ 2\ 3)$, and $\alpha\beta\alpha = (2\ 3)$. Since $\alpha^2 = \beta^2 = 1$ is not enough to determine the order of $\alpha\beta$ (because $\alpha\beta \neq \beta\alpha$), then we must add $(\alpha\beta)^2 = 1$. Then we have the presentation $S_3 = \langle \alpha, \beta \mid \alpha^2 = \beta^2 = (\alpha\beta)^3 = 1 \rangle$. $\square$

1.4 Matrix Groups

Let F be a field and let $n \in \mathbb{Z}^+$.

(1) Prove that $|\mathrm{GL}_2(\mathbb{F}_2)| = 6$.

Solution. Note that $\mathbb{F}_2 = \{0, 1\}$. Moreover, consider

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{F}_2)$$

The determinant is $ad - bc$, which is never 0 whenever a, d are nonzero or when b, c are nonzero, but not both. Then

$$\mathrm{GL}_2(\mathbb{F}_2) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

$\square$

(2) Write out all the elements of $\mathrm{GL}_2(\mathbb{F}_2)$ and compute the order of each element.

Solution. By direct calcuation, we obtain the following:

A	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$
$\lvert A \rvert$	1	2	2	2	3	3

$\square$

(3) Show that $\mathrm{GL}_2(\mathbb{F}_2)$ is non-abelian.

Solution.

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

$\square$

(4) Show that if n is not prime then $\mathbb{Z}/n\mathbb{Z}$ is not a field.

Solution. Since n is not prime, there exists $a, b \in \mathbb{Z}^+$ such that $n = ab$. Since $(a, n) \neq 1$, then Section 0, Proposition 4 says that a has no multiplicative inverse. $\square$

(5) Show that $\mathrm{GL}_n(F)$ is a finite group if and only if F has a finite number of elements.

Solution. Suppose F is infinite. Note that the set $\{\alpha I \mid \alpha \in F^\times\} \subseteq \mathrm{GL}_n(F)$. Then $\mathrm{GL}_n(F)$ is infinite.

Suppose $|F| = m < \infty$. Then there are m^{n^2} potential $n \times n$ matrices, and less whose determinants are nonzero. Then $|\mathrm{GL}_n(F)|$ is finite. $\square$

(6) If $|F| = q$ is finite prove that $|\mathrm{GL}_n(F)| < q^{n^2}$.

Solution. See previous solution. $\square$

(7) Let p be a prime. Prove that the order of $\mathrm{GL}_2(\mathbb{F}_p)$ is $p^4 - p^3 - p^2 + p$ (do not just quote the order formula in this section). [Subtract the number of 2×2 matrices which are *not* invertible from the total number of 2×2 matrices over $\mathbb{F}_p$. You may use the fact that a 2×2 matrix is not invertible if and only if one row is a multiple of the other.]

Solution. Note that the number of 2×2 matrices with entries from $\mathbb{F}_p$ is p^4. Using the given fact, let r_1 denote the first row and r_2 the second row of the matrices. Then if $r_1 = kr_2$ for some $k \in \mathbb{F}_p$, then the matrix is linearly independent. We then cover the cases:

1. Suppose $r_1 = 0$. Then r_2 is automatically a multiple of r_1, and there are p^2 choices for r_2 since there are 2 entries.
2. Suppose $r_1 \neq 0$. Then we have $p^2 - 1$ choices for r_1. Since there are p scalars, then we have p choices for r_2 (take kr_1 for $k \in \{0, 1, \ldots, p-1\}$. Then there are $(p^2-1)p = p^3 - p$ matrices that are not invertible.

The total of matrices that are not invertible is then $p^3 + p^2 - p$. The result follows. $\square$

(8) Show that $\mathrm{GL}_n(F)$ is non-abelian for any $n \geq 2$ and any F.

Solution. Note that every F has elements 0 and 1 as additive and mutliplicative identities respectively, so we may safely assume $\mathbb{F}_2$ to be our field.

Proceeding by induction, the case of $n = 2$ was proven in Exercise 3. Now suppose $\mathrm{GL}_{n-1}(\mathbb{F}_2)$ is non-abelian for $n \geq 3$. Let $A, B \in \mathrm{GL}_{n-1}(\mathbb{F}_2)$ be noncommuting matrices. Using a block matrix, then

$$\begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} B & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} AB & 0 \\ 0 & 1 \end{pmatrix} \neq \begin{pmatrix} BA & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} B & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}$$

$\square$

(9) Prove that the binary operation of matrix multiplication of 2×2 matrices with real number entries is associative.

Solution.

$$\begin{aligned}
\left[\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}\right]\begin{pmatrix} a'' & b'' \\ c'' & d'' \end{pmatrix} &= \begin{pmatrix} aa' + bc' & ab' + bd' \\ ca' + dc' & cb' + dd' \end{pmatrix}\begin{pmatrix} a'' & b'' \\ c'' & d'' \end{pmatrix} \\
&= \begin{pmatrix} (aa' + bc')a'' + (ab' + bd')c'' & (aa' + bc')b'' + (ab' + bd')d'' \\ (ca' + dc')a'' + (cb' + dd')c' & (ca' + dc')b'' + (cb' + dd')d'' \end{pmatrix} \\
&= \begin{pmatrix} aa'a'' + bca'' + ab'c'' + bd'c'' & aa'b'' + bc'b'' + ab'd'' + bd'd'' \\ ca'a'' + dc'a'' + cb'c' + dd'c' & ca'b'' + dc'b'' + cb'd'' + dd'd'' \end{pmatrix} \\
&= \begin{pmatrix} a(a'a'' + b'c'') + b(c'a'' + d'c'') & a(a'b'' + b'd'') + b(c'b'' + d'd'') \\ c(a'a'' + b'c'') + d(c'a'' + d'c'') & c(a'b'' + b'd'') + d(c'b'' + d'd'') \end{pmatrix} \\
&= \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} a'a'' + b'c'' & a'b'' + b'd'' \\ c'a'' + d'c'' & c'b'' + d'd'' \end{pmatrix} \\
&= \begin{pmatrix} a & b \\ c & d \end{pmatrix}\left[\begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}\begin{pmatrix} a'' & b'' \\ c'' & d'' \end{pmatrix}\right]
\end{aligned}$$

$\square$

(10) Let

$$G = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \,\middle|\, a, b, c \in \mathbb{R}, a \neq 0, c \neq 0 \right\}$$

(a) Compute the product of $\begin{pmatrix} a_1 & b_1 \\ 0 & c_1 \end{pmatrix}$ and $\begin{pmatrix} a_2 & b_2 \\ 0 & c_2 \end{pmatrix}$ to show that G is closed under matrix multiplication.

(b) Find the matrix inverse of $\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$ and deduce that G is closed under inverses.

(c) Deduce that G is a subgroup of $GL_2(\mathbb{R})$ (cf. Exercise 26, Section 1).

(d) Prove that the set of elements of G whose two diagonal entries are equal (i.e., $a = c$) is also a subgroup of $GL_2(\mathbb{R})$.

Solution.

(a)

$$\begin{pmatrix} a_1 & b_1 \\ 0 & c_1 \end{pmatrix}\begin{pmatrix} a_2 & b_2 \\ 0 & c_2 \end{pmatrix} = \begin{pmatrix} a_1a_2 & a_1b_2 + b_1c_2 \\ 0 & c_1c_2 \end{pmatrix}$$

Since $a_i \neq 0$ and $c_i \neq 0$, then $a_1a_2 \neq 0$ and $c_1c_2 \neq 0$ so that G is closed under matrix multiplication.

(b) The determinant is ac. We then have the inverse

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}^{-1} = \frac{1}{ac}\begin{pmatrix} c & -b \\ 0 & a \end{pmatrix} = \begin{pmatrix} 1/a & -b/ac \\ 0 & 1/c \end{pmatrix}$$

(c) Since G is closed under inverses and matrix multiplication, the result follows.

(d) Let $H = \{A \in G \mid a = c\}$. Then

$$\begin{pmatrix} a_1 & b_1 \\ 0 & a_1 \end{pmatrix}\begin{pmatrix} a_2 & b_2 \\ 0 & a_2 \end{pmatrix} = \begin{pmatrix} a_1a_2 & a_1b_2 + b_1a_2 \\ 0 & a_1a_2 \end{pmatrix} \in H$$

and

$$\begin{pmatrix} a & b \\ 0 & a \end{pmatrix}^{-1} = \frac{1}{a^2}\begin{pmatrix} a & -b \\ 0 & a \end{pmatrix} = \begin{pmatrix} 1/a & -b/a^2 \\ 0 & 1/a \end{pmatrix} \in H$$

Then H is closed under inverses and matrix multiplication, hence it is a subgroup of $GL_2(\mathbb{R})$. □

(11) Let

$$H(F) = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \middle| \, a, b, c \in F \right\}$$

called the *Heisenberg group* over F. Let

$$X = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \text{ and } Y = \begin{pmatrix} 1 & d & e \\ 0 & 1 & f \\ 0 & 0 & 1 \end{pmatrix}$$

be elements of $H(F)$.

(a) Compute the matrix product XY and deduce that $H(F)$ is closed under matrix multiplication. Exhibit explicit matrices such that $XY \neq YX$ (so that $H(F)$ is always non-abelian).

(b) Find an explicit formula for the matrix inverse X^{-1} and deduce that $H(F)$ is closed under inverses.

(c) Prove the associative law for $H(F)$ and deduce that $H(F)$ is a group of order $|F|^3$. (Do not assume that matrix multiplication is associative.)

(d) Find the order of each element of the finite group $H(\mathbb{Z}/2\mathbb{Z})$.

(e) Prove that every nonidentity of the group $H(\mathbb{R})$ has infinite order.

Solution.

(a) The product is calculated:

$$XY = \begin{pmatrix} 1 & d+a & e+af+b \\ 0 & 1 & f+c \\ 0 & 0 & 1 \end{pmatrix} \in H(F)$$

Moreover, we have that

$$\begin{pmatrix}1&1&0\\0&1&0\\0&0&1\end{pmatrix}\begin{pmatrix}1&0&0\\0&1&1\\0&0&1\end{pmatrix}=\begin{pmatrix}1&1&1\\0&1&1\\0&0&1\end{pmatrix}\neq\begin{pmatrix}1&1&0\\0&1&1\\0&0&1\end{pmatrix}=\begin{pmatrix}1&0&0\\0&1&1\\0&0&1\end{pmatrix}\begin{pmatrix}1&1&0\\0&1&0\\0&0&1\end{pmatrix}$$

(b) To calculate the inverse, one can use Gaussian elimination:

$$\left(\begin{array}{ccc|ccc}1&a&b&1&0&0\\0&1&c&0&1&0\\0&0&1&0&0&1\end{array}\right)\implies\left(\begin{array}{ccc|ccc}1&0&0&1&-a&ac-b\\0&1&0&0&1&-c\\0&0&1&0&0&1\end{array}\right)$$

The right 3×3 matrix, call it Z, is the inverse of X and can be shown as such through direct calculation, i.e., $XZ=ZX=I$ so that $X^{-1}=Z$.

(c)

$$\begin{aligned}\left[\begin{pmatrix}1&a&b\\0&1&c\\0&0&1\end{pmatrix}\begin{pmatrix}1&d&e\\0&1&f\\0&0&1\end{pmatrix}\right]\begin{pmatrix}1&g&h\\0&1&i\\0&0&1\end{pmatrix}&=\begin{pmatrix}1&a+d&af+b+e\\0&1&c+f\\0&0&1\end{pmatrix}\begin{pmatrix}1&g&h\\0&1&i\\0&0&1\end{pmatrix}\\&=\begin{pmatrix}1&a+d+g&h+i(a+d)+af+b+e\\0&1&c+f+i\\0&0&1\end{pmatrix}\\&=\begin{pmatrix}1&a+d+g&ai+ad+af+b+e+h\\0&1&c+f+i\\0&0&1\end{pmatrix}\\&=\begin{pmatrix}1&a&b\\0&1&c\\0&0&1\end{pmatrix}\begin{pmatrix}1&d+g&di+e+h\\0&1&f+i\\0&0&1\end{pmatrix}\\&=\begin{pmatrix}1&a&b\\0&1&c\\0&0&1\end{pmatrix}\left[\begin{pmatrix}1&d&e\\0&1&f\\0&0&1\end{pmatrix}\begin{pmatrix}1&g&h\\0&1&i\\0&0&1\end{pmatrix}\right]\end{aligned}$$

Moreover, there are 3 choices to vary in each matrix in $H(F)$, such as a,b,c in X. Then there are $|F|^3$ elements in $H(F)$.

(d) To save time, denote a matrix in $H(\mathbb{Z}/2\mathbb{Z})$ as the 3-tuple (a,b,c), where regular matrix multiplication is $(a,b,c)(d,e,f)=(a+d,af+b+e,c+f)$. Note that $(a,b,c)^2=(2a,ac+2b,2c)$. Since $a,b,c\in\mathbb{Z}/2\mathbb{Z}$, this degrades to $(0,ac,0)$. We then have two cases to consider:

1. Suppose $ac=0$. Then at least one of a and c is 0, while b can be 0 or 1. Then the matrices are $(0,0,0),(1,0,0),(0,0,1),(0,1,0),(1,1,0),(0,1,1)$. Except for $(0,0,0)$ which is the identity matrix and thus has order 2, squaring the other matrices results in $(0,0,0)$ so that each have order 2.
2. Suppose $ac=1$. Then this occurs when both $a=c=1$ and b is either 0 or 1. These are the matrices $(1,1,0)$ and $(1,1,1)$. Since squaring these matrices results in $(0,0,1)$ which has order 2. then the aforementioned matrices have order 4.

(e) Utilizing the same formulation as in part (d), let $A=(a,b,c)\in H(\mathbb{R})$. We prove that

$$(a,b,c)^n=(na,n(n-1)ac/2+nb,nc)$$

To that end, it is clear that $n=1$ holds true. Suppose it holds for some n. Then for $n+1$, we have

$$\begin{aligned}(a,b,c)^{n+1}&=(a,b,c)^n(a,b,c)\\&=(na,n(n-1)ac/2+nb,nc)(a,b,c)\\&=(na+a,(na)c+n(n-1)ac/2+nb+b,nc+c)\\&=((n+1)a,n(n+1)ac/2+(n+1)b,(n+1)c)\end{aligned}$$

so that the relationship is true by induction. Since $A \neq I$, then at least one of a, b, c must be nonzero. Then for any $n \in \mathbb{Z}^+$, none of the integers na, $n(n-1)ac/2 + nb$, or nc are 0. Then no power of (a, b, c) results in $(0,0,0)$, hence no nonidentity element of $H(\mathbb{R})$ has any finite order. Therefore, every nonidentity element of $H(\mathbb{R})$ has infinite order. □

1.5 The Quaternion Group

(1) Compute the order of each of the elements in Q_8.

Solution.

a	1	-1	i	$-i$	j	$-j$	k	$-k$
$\lvert a\rvert$	1	2	4	4	4	4	4	4

□

(2) Write out the group tables for S_3, D_8, and Q_8.

Solution. S_3:

$\circ$	1	(1 2)	(1 3)	(2 3)	(1 2 3)	(1 3 2)
1	1	(1 2)	(1 3)	(2 3)	(1 2 3)	(1 3 2)
(1 2)	(1 2)	1	(1 2 3)	(1 3 2)	(1 3)	(2 3)
(1 3)	(1 3)	(1 3 2)	1	(1 2 3)	(2 3)	(1 2)
(2 3)	(2 3)	(1 2 3)	(1 3 2)	1	(1 2)	(1 3)
(1 2 3)	(1 2 3)	(2 3)	(1 2)	(1 3)	(1 3 2)	1
(1 3 2)	(1 3 2)	(1 3)	(2 3)	(1 2)	1	(1 2 3)

D_8:

$\circ$	1	r	r^2	r^3	s	sr	sr^2	sr^3
1	1	r	r^2	r^3	s	sr	sr^2	sr^3
r	r	r^2	r^3	1	sr^3	s	sr	sr^2
r^2	r^2	r^3	1	r	sr^2	sr^3	s	sr
r^3	r^3	1	r	r^2	sr	sr^2	sr^3	s
s	s	sr	sr^2	sr^3	1	r	r^2	r^3
sr	sr	sr^2	sr^3	s	r^3	1	r	r^2
sr^2	sr^2	sr^3	s	sr	r^2	r^3	1	r
sr^3	sr^3	s	sr	sr^2	r	r^2	r^3	1

Q_8:

$\cdot$	1	-1	i	$-i$	j	$-j$	k	$-k$
1	1	-1	i	$-i$	j	$-j$	k	$-k$
-1	-1	1	$-i$	i	$-j$	j	$-k$	k
i	i	$-i$	-1	1	k	$-k$	j	$-j$
$-i$	$-i$	i	1	-1	$-k$	k	$-j$	j
j	j	$-j$	$-k$	k	-1	1	i	$-i$
$-j$	$-j$	j	k	$-k$	1	-1	$-i$	i
k	k	$-k$	j	$-j$	$-i$	i	-1	1
$-k$	$-k$	k	$-j$	j	i	$-i$	1	-1

□

(3) Find a set of generators and relations for Q_8.

Solution. An extra relation is necessary to intertwine i, j, k together, so a presentation for Q_8 is:

$$Q_8 = \langle -1, i, j, k \mid i^2 = j^2 = k^2 = ijk = -1 \rangle$$

□